

CARDIOVASCULAR DISEASES

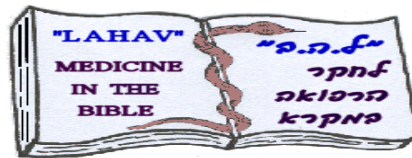


*Medical Research in Biblical Times
from the Viewpoint
of Contemporary Perspective*

Liubov Ben-Nun



Ben-Gurion University of the Negev
Liubov Ben-Nun
Professor Emeritus
Faculty of Health Sciences, Dept. of Family Medicine
Beer-Sheva, Israel



This Book describes medical situations associated with cardiovascular diseases.

These situations include:

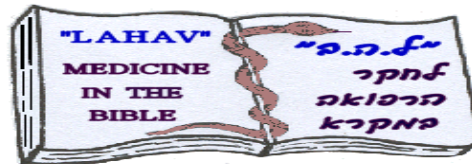
Peripheral Vascular Disease, Syncope, Acute Myocardial Infarction, and Resuscitation.

NOT FOR SALE

CARDIOVASCULAR DISEASES



Ben-Gurion University of the Negev
Professor Emeritus Liubov Ben-Nun
Faculty of Health Sciences, Dept. of Family Medicine
Beer-Sheva, Israel



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Fax: +(972) 8 6883376 Mobile 050 5971592
E-Mail: L-bennun@smile.net.il

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PREFACE

The purpose of this research is to analyze the medical situations and conditions referred to in the Bible, as we are dealing with a contemporary medical record.

These are scientific medical studies incorporating verses from the Bible, without no interpretation or historical descriptions of places.

Fundamentally, this Research is constructed purely from an examination of passages from the Bible, exactly as written.

The research is part of a long series of published studies on the subject of biblical medicine from a modern medical perspective.

This is not a laboratory research. The Research is built entirely on a secular foundation. With due to respects to people faith, this Research takes a modern look at medical practices. Each to his own beliefs.

FOREWORD

In Ancient Egypt, the cardiovascular complaints take up a significant place in the medical papyri, which inform us about the medical knowledge. The heart was depicted circa 3000 BC with remarkable accuracy. The Ebers papyrus contains the first Essay on Cardiology of Humanity. The Egyptian physicians were the pioneers of the rigorous cardiovascular examination. They worked out an innovative design of cardiovascular anatomy and physiology, which influenced all the civilizations of the Mediterranean Basin that lasted more than thirty centuries (1). The heart was regarded in Ancient Egypt as the organic motor of the body and the seat of intelligence, an important religious and spiritual symbol. It was considered as one of the eight parts of human body. Counter to other organs it had to be kept carefully intact in the mummy to ensure its eternal life. The concept of heart included three constituents: heart-haty, heart-ib, and the spiritual seat of intelligence, emotion and memory. The hieroglyphs representing the heart early in the first dynasty were drawn with eight vessels attached to it. Egyptian doctors have elaborated an original conception of cardiovascular physiology, which endured 30 centuries (2).

In the 6th century BCE, the Ayurvedic physician Sushruta in ancient India knew the knowledge of circulation of vital fluids through the body. He also seems to have possessed knowledge of the arteries (3). A physician of the Hippocratic School around the 4th century BCE discovered the valves of the heart. However, their function was not properly understood because blood pools in the veins and arteries after death look empty. Ancient anatomists assumed they were filled with air and were for transport of air.

In 2nd century AD Rome, the Greek physician Galen knew that blood vessels carried blood and identified venous (dark red) and arterial (brighter and thinner) blood, each with distinct and separate functions. Growth and energy were derived from venous blood created in the liver from chyle, while arterial blood gave vitality by containing pneuma (air) and originated in the heart. Blood flowed from both creating organs to all parts of the body where it was consumed and there was no return of blood to the heart or liver. The heart did not pump blood around, the heart's motion sucked blood during diastole and the blood moved by the pulsation of the arteries themselves (6).

Galen believed that the arterial blood was created by venous blood passing from the left ventricle to the right by passing through

'pores' in the interventricular septum, while air passed from the lungs via the pulmonary artery to the left side of the heart. As the arterial blood was created 'sooty' vapors were created and passed to the lungs also via the pulmonary artery to be exhaled (6).

In 1025, *The Canon of Medicine* by the Persian physician, Avicenna, "erroneously accepted the Greek notion regarding the existence of a hole in the ventricular septum by which the blood traveled between the ventricles." Despite this, Avicenna "correctly wrote on the cardiac cycles and valvular function", and "had a vision of blood circulation" in his *Treatise on Pulse* (4). While also refining Galen's erroneous theory of the pulse, Avicenna provided the first correct explanation of pulsation: "Every beat of the pulse comprises two movements and two pauses. Thus, expansion: pause: contraction: pause. The pulse is a movement in the heart and arteries ... which takes the form of alternate expansion and contraction" (5).

In 1242, the Arabian physician, Ibn al-Nafis, became the first person to accurately describe the process of pulmonary circulation, for which he is sometimes considered as the father of circulatory physiology (6). Ibn al-Nafis stated in his *Commentary on Anatomy in Avicenna's Canon*: "...the blood from the right chamber of the heart must arrive at the left chamber but there is no direct pathway between them. The thick septum of the heart is not perforated and does not have visible pores as some people thought or invisible pores as Galen thought. The blood from the right chamber must flow through the vena arteriosa (pulmonary artery) to the lungs, spread through its substances, be mingled there with air, pass through the arteria venosa (pulmonary vein) to reach the left chamber of the heart and there form the vital spirit..." (6).

In addition, Ibn al-Nafis had an insight into what would become a larger theory of the capillary circulation. He stated that "there must be small communications or pores (manafidh in Arabic) between the pulmonary artery and vein," a prediction that preceded the discovery of the capillary system by more than 400 years (7). Ibn al-Nafis' theory, however, was confined to blood transit in the lungs and did not extend to the entire body.

Finally William Harvey, a pupil of Hieronymus Fabricius (who had earlier described the valves of the veins without recognizing their function), performed a sequence of experiments, and published *Exercitatio Anatomica de Motu Cordis et Sanguinis in Animalibus* in 1628, which "demonstrated that there had to be a direct connection between the venous and arterial systems throughout the body, and not just the lungs. Most importantly, he argued that the beat of the heart

produced a continuous circulation of blood through minute connections at the extremities of the body. This is a conceptual leap that was quite different from Ibn al-Nafis' refinement of the anatomy and blood flow in the heart and lungs" (8). This work, with its essentially correct exposition, convinced slowly the medical world. However, Harvey was not able to identify the capillary system connecting arteries and veins; these were later discovered by Marcello Malpighi in 1661.

Numerous physicians have studied the anatomy and physiology of the cardiovascular system. These studies are important since cardiovascular diseases have afflicted humans since the dawn of history. Even today, many cardiovascular diseases have not disappeared from the scene. The descriptions of various cardiovascular diseases give us a picture of how these diseases were perceived in ancient times, where the diseases occurred, the likely diagnosis, the consequences of cardiovascular diseases, and how humans coped with them. Evaluation of possible cases of the cardiovascular diseases described in the Bible may increase our understanding of the epidemiology, the clinical picture and ways of handling these diseases in modern times.

A variety of factors affects cardiovascular diseases. In order to better understand our future we should study our past. Study of the past provides an insight into the different aspects of various cardiovascular diseases such as peripheral vascular disease, syncope, acute myocardial infarction and the principles of resuscitation. A comprehensive understanding of available information on the history of cardiovascular diseases can widen the horizon of our knowledge and help us to deal more comprehensively with these diseases.

The experience of a diagnostic challenge with several treatment possibilities, and analysis of towards issues associated with patients afflicted by cardiovascular diseases give us an excellent opportunity to examine matters from the past that are very relevant to modern times.

All biblical texts were examined and medical situations related to cardiovascular diseases were studied closely. The importance of this book is that it provides tools for a better comprehension of the nature of cardiovascular disease drawn from logical conclusions concerning diseases that affected people in biblical times. It gives health care providers an excellent opportunity to re-examine and re-evaluate the clinical manifestations of cardiovascular diseases and in some cases the therapeutic approaches towards these diseases.

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CARDIOVASCULAR DISEASES

PERIPHERAL VASCULAR DISEASE

Introduction

Patients have suffered from pain in their legs, caused by various musculoskeletal, vascular, neurological, metabolic and traumatic diseases, for thousands of years. Pain may be due to local diseases or abnormalities in legs, or can be referred from other parts of body. Who suffered from leg pains in biblical times? What are the characteristics of the disease of the legs? What is the most likely diagnosis? This article aims to answer these questions by evaluating a disease of the legs as described in the Bible.

Disease of Legs as Described in the Bible

King Asa, the third of the King of the house of Judah, reigned between 867-906 BCE. When he reached old age the King suffered

from a disease of the legs. “*Nevertheless in the time of his old age he was diseased in his legs*” (I Kings 15:23). Thus, it is clearly stated that the King suffered from a disease which involved his legs.

When did the King’s disease develop? The disease developed “*.. in the time of his old age...*”, that is, the King was already old when the disease attacked him. Did the disease damage one or both legs? Since King Asa “*...was diseased in his legs,*” it follows that the disease involved both his legs.

Legs include thighs, knee, calves, ankles, and feet. What was the disease that afflicted the elderly King Asa? Where was the disease located in the King’s legs?

Referred pain

The term “sciatica” is applied to pain radiating from the buttocks, down the back of the thigh, and along the posterior or lateral aspect of the calf to the foot. The causes of sciatic pain include the affections of the nerve roots, lumbosacral plexus and sciatic nerve: *cauda equina* - neurofibroma or other tumor, backward protrusion of intervertebral disc, irritation of meninges by hemorrhage, infection and intrathecal injections, hydatid cysts, postherpetic neuralgia; *lumbar vertebrae*- disk lesions, spondylosis, Pott’s disease, osteomyelitis, tumors, fracture-dislocations, spondylolisthesis; *lumbo-sacral plexus* - tumor of rectum; and *sciatic nerve* - neurofibroma, and penetrating injuries. *Extraneural* causes include *sacro-iliac joint* - subluxation, tuberculous and non-tubercular arthritis, ankylosing spondylitis and other spondylarthropathies; *sacrum and pelvic bones* - primary and secondary neoplasms; and *soft tissues* - gluteal bursitis. Although there are numerous causes of sciatica, the vast majority of all cases are due to lesions of the lumbar intervertebral disc, spondylosis or sacroiliac disease (1).

Did King Asa suffer from referred pain in his legs? Since the King “*...was diseased in his legs*” it is most likely that the disease developed in his legs per se, and not in other parts of his body. Thus, all diagnoses related to referred pain can be excluded.

Pain in the Thigh

Pain down the front of the thigh to the knee may be referred from the hip, as in tuberculosis and osteoarthritis of that joint, from the upper part of the iliac bone, or from the 2nd or 3rd lumbar vertebra with protrusion of lumbar disks. Some pains are due to diabetes, Pott’s disease, or disease of meninges and cauda equina. Obturator pain, down the inner aspect of the thigh, is commonly due to disease

of the hip joint, rarely to obturator hernia or gross neoplastic or inflammatory disease within the pelvis. The lateral cutaneous nerve of the thigh is sometimes compressed as it passes from the pelvis under the lateral aspect of inguinal ligament where it may become involved in the insertion of the sartorius muscle. This compression is associated with pain and numbness over the lateral aspect of the thigh (1). Did King Asa suffer from a disease of the thigh? Since the disease developed in both legs, it is unlikely that only the thighs were involved. Thus, all diseases related only to the thigh(s) can be excluded.

Knee Pain

Acute knee pain is related to synovitis (e.g., rheumatoid arthritis, gout, pseudogout, Reiter's syndrome, rheumatic fever, viral arthritis, systemic lupus erythematosus, septic arthritis), intermittent hydrarthrosis, Baker's cyst, acute anserine, prepatellar, or infrapatellar bursitis, subluxation of the patella, Osgood-Schlatter disease, discoid meniscus, osteochondritis dissecans, spontaneous osteonecrosis of the knee, tumor, traumatic hemarthrosis, ligamentous sprains or tears, meniscal tears, and fracture of the tibial plateau or patella. Chronic knee pain is associated with degenerative osteoarthritis, genu valgum, varum, chronic infection (e.g., tuberculosis), chondromalacia of the patella, tumor of bone or synovium, synovitis, and torn meniscus (2).



Did the King suffer from any disease listed above in his knee? Since it is stated that the King “...*was diseased in his legs*,” all diagnoses associated with only knee diseases, either acute or chronic, can be excluded.

Foot Problems

There are various medical conditions which may affect the feet: *dermatological* - hyperkeratosis, dermal ulceration, corns, calluses, warts, tinea pedis, tinea unguium, incurvated, ingrown, atrophic, or hypertrophic nails, psoriasis, and neoplastic diseases; *musculoskeletal*- exostoses, hallux valgus, hammertoe, mallet toe, other pedal degenerative joint disease, bone deformity, pes planus or pes cavus, overlapping, rotated digits, heel spur, plantar fasciitis, ankle equinus, muscle or joint dysfunction, limb length inequality, calcificated Achilles tendonitis bursitis or spontaneous rupture of the Achilles tendon and metatarsalgia; *neurological* - neuropathies, tarsal tunnel syndrome, Morton's neuroma, foot dysfunction secondary to gait abnormalities due to Parkinson's disease or multiple sclerosis, and atrophic or spastic conditions of the lower extremities; *vascular* - arteriosclerosis obliterans, microangiopathy, microembolization, distal tibial vascular insufficiency, venous stasis/ulceration, phlebitis/postphlebotic syndrome, Raynaud disease/phenomenon; and *other* calcaneodynia due to Reiter's disease, march fracture, erythromelalgia found in the early stages of chronic arsenical poisoning, polycythemia, and syringomyelia (1,3-6).



Calcified Achilles tendonitis

Did the King suffer from any condition listed above? Since the King “*..was diseased in his legs,*” we assume the disease affected his legs. Thus, all diagnoses associated with only foot diseases can be excluded.

Tabes Dorsalis

The lightning pains of tabes occur more often in the legs than in other parts of the body. Unlike neuralgia, they are usually bilateral and not refer to the distribution of any particular peripheral nerve. Whether trivial and “niggling,” or so intense as to draw sweat and groans from the most heroic of sufferers, they are always short and

lightning-like in duration, often repeated in paroxysms, irregularly periodic in their attacks, and variable in their localization (1). In this case, important factors are history of syphilis, failure of the pupils to react to light but reacting on convergence, absence of knee and ankle reflexes, loss of sensitivity to pain below the knees and over the trunk and inner arms, impaired postural sense in the toes, lymphocytosis in the cerebrospinal fluid, and positive TPI and VDRL tests in the blood and cerebrospinal fluid (though they may be negative) (1).

Was this disease responsible for the disease in the King's legs? In the absence of anamnestic, physical and laboratory findings, this diagnosis seems very unlikely.

Degenerative Joint Disease

Degenerative joint disease is the single most common form of arthritis afflicting humankind, and severe forms of this disease account for high rates of disability in the elderly (7). Joints characteristically affected by osteoarthritis are hips, knees, distal and proximal interphalangeal joints of fingers, the first carpometacarpal joint, first metatarsophalangeal joints of feet (hallux rigidus), shoulders, cervical spine, and lumbar spine (7,8).

Did King Asa suffer from degenerative osteoarthritis affecting the knees and the first metatarsophalangeal joints? Since the disease damaged the King's legs, it is unlikely that only knees, and/or the first metatarsophalangeal joints were affected.

Rheumatoid Arthritis

Rheumatoid arthritis begins most frequently as a symmetric peripheral polysynovitis with warmth and swelling of many peripheral joints, including knees, ankles, elbows, shoulders, wrists and the small joints of the hands and feet. In the fingers, the metacarpophalangeal joints and in the toes, the metatarsal joints are frequently involved. Arthritis is accompanied by constitutional manifestations such as malaise, anorexia, low-grade fever, weight loss, and weakness (8).

Did King Asa suffer from rheumatoid arthritis affecting his knees and the small feet joints? It seems very unlikely that rheumatoid arthritis affected only his knees and small feet joints. Also, no additional constitutional symptoms are mentioned in the Biblical text. For these reasons, this diagnosis is very unlikely.

Gout

Gout was well described clinically by Greek and Roman physicians, but it was not until the latter part of the nineteenth century that crystals of monosodium urate were observed in the joint fluid during acute gouty attacks and not until the 1960's that these crystals were recognized as the cause of the attack (7). About 90 percent of first attacks occur in a single joint, and at least half in the first metatarsal phalangeal joint. Following the toe in order of frequency as sites of initial involvement are the insteps, ankles, heels, knees, wrists, fingers, and elbows (9). Acute gouty arthritis may progress to a stage of chronic polyarthropathy (8).

Did King Asa suffer from an acute attack of gout? The occurrence of acute gouty arthritis is very unlikely since the disease usually involves a single joint, and most probably the first metatarsal phalangeal joint. Since the King “*..was diseased in his legs*” an acute attack of gout with the involvement of a single joint is very unlikely. Did acute gout progress into chronic polyarthropathy? It can be speculated that King Asa may have suffered from chronic gouty arthropathy involving the joints in his legs (10). However, if this diagnosis is accepted, there is no explanation for the disease, which affected the King's thighs and calves. Thus, this diagnosis does not entirely explain what kind of disease involved the *whole* of the King's legs.



Figure 1



Figure 2

Polyneuropathy

Polyneuropathy denotes a widespread symmetric process, usually distal and graded. Polyneuropathies are associated with various systemic diseases including diabetes mellitus, uremia, porphyria, hypoglycemia, vitamin B12 deficiency, chronic liver

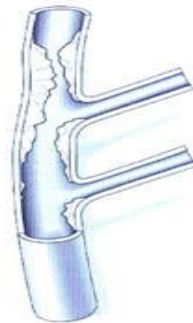
disease, chronic obstructive lung disease, carcinoma, multiple myeloma, lymphoma, malabsorption, drugs and environmental toxins, hereditary disorders, and infectious diseases. Idiopathic polyneuropathies can also be encountered (11).

Did King Asa suffer from any of the diseases mentioned above with the involvement of his legs? Although peripheral neuropathy can occur, the absence of additional signs, for example, weakness, polyuria, polydipsia, bone, or abdominal pain makes these diagnoses very unlikely.

Peripheral Artery Disease

Peripheral Artery Disease (PAD), usually caused by atherosclerosis, is defined as an obstructive arterial disease of the lower extremities that reduces arterial flow during exercise or, in advanced stages, at rest (12). PAD represents a marker for premature cardiovascular events, and in patients with PAD, even in the absence of a history of myocardial infarction (MI) or ischemic stroke; they have approximately the same relative risk of death from cardiovascular causes as do patients with a history of coronary or cerebrovascular disease (13).

Did King Asa suffer from PAD? Since “*..in the time of his old age he was diseased in his legs,*” it is very likely that PAD afflicted the King. The disease developed gradually in the elderly King Asa’s legs affecting his thighs and calves.



Peripheral vascular disease

The Prevalence

PAD is a common disorder in the general population. The prevalence increases with age and is under the influence of vascular

risk factors (14-21). It is a progressive atherosclerotic condition affecting approximately 27 million people in North America and Europe (22), and more than 8.5 million people in the USA (12). Albeit a common clinical malady, it is underrecognized and undertreated (22).

Obstructive arterial disease is usually secondary to arteriosclerosis and most frequently common in elderly males. This disease accounts for at least 95 per cent of the cases of chronic ischemia of the legs (23,24). In addition, there is an increased prevalence of peripheral atherosclerotic occlusive disease in individuals with hypertension, diabetes, hypercholesterolemia, and cigarette smoking (23).

The prevalence of PAD in 6979 men and women, aged ≥ 50 years with a history of cigarette smoking or diabetes mellitus, screened for PAD by an ankle-brachial index (ABI) is 29% (18). Among these patients with PAD, classic claudication is present in 11%. While the prevalence of symptomatic PAD is 32% in 1160 men, mean age 80 years, and 26% in 2464 women, mean age 81 year, living in a nursing home (19).



Peripheral Vascular Disease

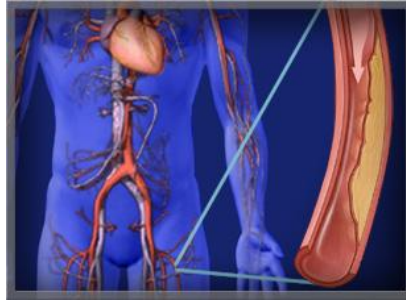
Symptomatology

PAD in the elderly can be: 1) asymptomatic, 2) associated with intermittent claudication, or 3) cause critical limb ischemia (12,21,25).

Intermittent claudication is pain or weakness when walking that is relieved with rest. The muscle pain or weakness after exercise occurs distal to the arterial obstruction. Since the superficial femoral and popliteal arteries are most commonly affected by atherosclerosis, the pain of intermittent claudication is most commonly localized in the calf. Atherosclerotic obstruction of the distal aorta and its bifurcation

into the two iliac arteries may cause pain in the buttocks, hips, thighs, or the inferior back muscles as well as the calves (25,26).

Disease severity varies from patients who are asymptomatic to those who have unremitting symptoms. A high overlap exists between PAD and coronary artery and cerebrovascular disease. Risks for long-term morbidity and mortality are identical for PAD, intermittent claudication, and coronary artery disease (CAD) (26).



Peripheral Vascular Disease

Physical Examination

The vascular physical examination includes (1) measurement of the blood pressure in both arms, (2) palpation of the carotid pulses and listening for carotid bruits, (3) auscultation of the abdomen and flank for bruits, (4) palpation of the abdomen and notation of the presence of the aortic pulsation and its maximal diameter, (5) palpation of pulses at the brachial, radial, ulnar, femoral, popliteal, dorsalis pedis, and posterior tibial sites, and (6) auscultation of both femoral arteries for femoral bruits (27).

The shoes and socks should be removed and the feet inspected. The color, temperature, and integrity of the skin should be evaluated and the presence of distal hair loss, trophic skin changes, hypertrophic nails, and ulcerations noted (27).

Noninvasive Diagnosis

Persons with PAD of the lower extremities have reduced or absent arterial pulses. Noninvasive tests used to assess lower extremity arterial blood flow include measurement of ankle and brachial artery systolic blood pressures, characterization of velocity wave form, and duplex ultrasonography. Measurements of ankle and brachial artery systolic blood pressures using a Doppler stethoscope

and blood pressure cuffs allows calculation of the ABI which is normally 0.9–1.2. An ABI of <0.90 is 95% sensitive and 99% specific for the diagnosis of PAD. The lower the ABI, the more severe the restriction of arterial blood flow, and the more serious the ischemia. ABIs of 0.6–0.9 usually correlate with mild to moderate intermittent claudication. ABIs of 0.4–0.6 usually correlate with severe intermittent claudication. With ABIs between 0.25 and 0.4, rest pain and tissue loss are often found. Patients with calcified arteries from diabetes mellitus or renal failure occasionally have relatively non-compressible arteries leading to falsely elevated ABI values in the normal range (28).

In addition to measuring arterial pressure in non-palpable arteries, Doppler ultrasound methods allow characterization of the flow versus time velocity waveform. Finding biphasic flow at the groin or monophasic flow more distally is evidence of arterial obstruction even when ABI measurements are falsely increased to normal levels because of calcification.

Duplex ultrasonography combines Doppler frequency measurements with two-dimensional images of blood vessels. The severity of flow restriction caused by an arterial stenosis can be accurately assessed by this most comprehensive noninvasive method (29). Duplex ultrasonography, computed tomographic angiography, and magnetic resonance angiography are useful in assessing the anatomic location and severity of PAD and in selecting suitable candidates for endovascular or surgical revascularization (27).

Treadmill exercise testing with and without pre-exercise and post exercise ABIs helps differentiate claudication from pseudoclaudication in patients with exertional leg symptoms (27). Treadmill exercise testing may be useful to diagnose PAD with a normal resting ABI but a reduced post exercise ABI. Treadmill exercise testing may objectively document the magnitude of symptom limitation in patients with claudication.

Risk Factors

Modifiable risk factors that predispose to PAD include cigarette smoking, diabetes mellitus, hypertension, dyslipidemia (17,20,30–35), increased plasma homocysteine levels (26,32,36), hypothyroidism (37), physical inactivity, and obesity (34).

Continuing smoking increases the risk of amputation in patients with intermittent claudication. Patency in lower extremity bypass grafts is also worse in smokers than in nonsmokers (38). Smoking cessation reduces the progression of PAD to critical leg ischemia and

reduces the risk of MI and death from vascular causes. Smoking cessation programs should be strongly encouraged in persons with PAD (39).

Did the King suffer from hypertension, diabetes, or hypothyroidism? Did he have increased plasma homocysteine levels, or dyslipidemia?

Did the King smoke? Incense from burning fragrant leaves and resin was already known as a ritual practice in Babylon and Egypt from the beginning of the 3rd millennium BCE. However, tobacco and its use for smoking were not known in the ancient world before the discovery of America. Nevertheless, among the natives of Northern and Central America tobacco and the burning of leaves and inhalation of their smoke were known from the 3rd century BCE, although cigarettes - cut tobacco wrapped in paper - only appeared in the 19th century CE (40).

Thus, it is obvious that King Asa did not smoke cigarettes, nor tobacco. Did he smoke leaves, branches and fragrant resins? This question remains open.

Coexistence with Other Atherosclerotic Diseases

PAD coexists with other atherosclerotic disorders (20,33,41-43). In a study of 1886 men and women, mean age 81 years, 270 of 468 persons (58%) with PAD had CAD and 159 of 468 persons (34%) with PAD had prior ischemic stroke (41). In a study of 1802 men and women, mean age 80 years, living in the community and seen in an academic geriatrics practice, 161 of 236 persons (68%) with PAD had coexistent CAD and 100 of 236 persons (42%) with PAD had coexistent prior ischemic stroke (42). In addition, in 1,006 men and women, mean age 72 years, if PAD was present, 63% had coexistent CAD, and 43% had prior ischemic stroke (20). In 273 patients, mean age 71 years, with CAD, the lower the ABI, the higher prevalence of 3-vessel or 4-vessel CAD (43).

Cardiovascular Mortality and Morbidity

Both symptomatic and asymptomatic subjects with PAD have increased mortality rates, mainly due to cardiovascular and cerebrovascular expressions of atherosclerotic disease (21).

Persons with PAD are at increased risk for all-cause mortality, cardiovascular mortality, and cardiovascular events (44-49). At 10-year follow-up of 565 men and women, mean age 66 years, PAD significantly increased the risk of all-cause mortality (relative risk [RR]=3.1), of mortality from cardiovascular disease (RR=5.9), and of

mortality from CAD (RR=6.6) (45). At 4-year follow-up of 1492 women, mean age 71 years, an ABI of 0.9 or less was associated with a RR of 3.1 for all-cause mortality after adjustment for age, smoking, and other risk factors (47).

Treatment

Persons with PAD must have proper foot care (27,50). They must wear properly fitted shoes. Careless nail clipping or injury from walking barefoot must be avoided. Feet should be washed daily and the skin kept moist with topical emollients to prevent cracks and fissures, which may have portals for bacterial infection. Fungal infection of the feet must be treated. Socks should be wool or other thick fabrics, and padding or shoe inserts may be used to prevent pressure sores.

Exercise rehabilitation programs have been documented to increase walking distance in persons with intermittent claudication through improvements in peripheral circulation, walking economy, and cardiopulmonary function (51). The optimal exercise program for improving claudication pain distance in persons with PAD uses intermittent walking to near-maximal pain during a program of at least 6 months (52). Strength training is less effective than treadmill walking (53). The ACC/AHA guidelines recommend a supervised exercise program for patients with intermittent claudication (27).

Supervised exercise training is recommended for a minimum of 30–45 minutes in sessions performed at least 3 times per week for a minimum of 12 weeks (27) and preferably for 6 months or longer (52). Among persons with PAD, self-directed walking exercise performed at least 3 times weekly is associated with significantly less functional decline during the subsequent year (54).

Risk factor modifications should include smoking cessation (55) as well as blood pressure and cholesterol management (22,25), antiplatelet therapy which is essential in all patients with PAD to prevent systemic atherosclerotic complications. Furthermore, for symptomatic relief exercise therapy is the main intervention while pharmacological treatment should be only complementary (21). Patients with PAD are physically impaired and have a higher risk of cardiovascular events; therefore, the treatment goals are aimed at decreasing their cardiovascular risk, as well as improving exercise and daily functional performance (12).

Antiplatelet drugs that decrease the incidence of vascular death, nonfatal MI, and nonfatal stroke in persons with PAD are aspirin, ticlodipine, and clopidogrel (56). Acetylsalicylic acid (ASA) is the

antiplatelet of choice, and clopidogrel should be used as an alternative if ASA therapy is contraindicated or intolerance is present (22).

Aspirin (ASA) plus dipyridamole is more efficacious than aspirin alone in the treatment of persons with PAD (56). Adverse hematologic effects associated with ticlodipine limit the use of this drug in the treatment of elderly persons with PAD (57).

In the Dutch Bypass Oral Anticoagulants or Aspirin Study, 2690 persons were randomized after infrainguinal bypass surgery to aspirin 80 mg daily or to oral anticoagulation with phenprocoumon or acenocoumarol to maintain an INR of 3.0–4.5 (BOA Study group 2000). At 21-month follow-up, there was insignificant difference between the two treatments in the primary outcome of infrainguinal graft occlusion. There was insignificant difference between the two treatments in the secondary outcomes of MI, stroke, amputation, or vascular death. However, persons treated with oral anticoagulant therapy had 1.96 times more major bleeding episodes than persons treated with oral aspirin. The American College of Cardiology (ACC)/American Heart Association (AHA) guidelines state that oral anticoagulant therapy with warfarin should not be given to reduce the risk of adverse cardiovascular ischemic events in persons with atherosclerotic lower extremity PAD (27).

An observational study was performed in 575 men and women, mean age 80 years, with symptomatic PAD and prior MI (58). Of the 575 persons, 85 persons (15%) had contraindications to the use of beta-blockers. Of the 490 persons without contraindications to the use of beta-blockers, 257 persons (52%) were treated with beta-blockers. Adverse effects causing cessation of beta-blockers occurred in 31 of the 257 persons (12%). At 32-month follow-up, use of beta-blockers caused a 53% significant independent decrease in the incidence of new coronary events in elderly persons with PAD and prior MI (58). Beta-blockers should be used to treat CAD in patients with PAD in the absence of contraindications to these drugs.

Statins have favorable effects on multiple interrelated aspects of vascular biology important in atherosclerosis. In particular, they have beneficial effects on inflammation, plaque stabilization, endothelial dysfunction, and thrombosis. Statins are beneficial in acute vascular events (12). In patients with PAD, the goal of LDL therapy should be to 70 mg/dL (34).

Persons with PAD should be treated with statins regardless of age, gender, or initial serum lipids levels (60). Treatment of dyslipidemia with statins reduces the incidence of mortality, cardiovascular events, and stroke in persons with PAD with and

without CAD (59,60), and improves exercise time until intermittent claudication (62).

At 5-year follow-up of 4444 men and women with CAD and hypercholesterolemia in the Scandinavian Simvastatin Survival Study, compared with placebo, simvastatin significantly decreased the incidence of intermittent claudication by 38% (59).

Hypertension should be adequately controlled to decrease cardiovascular mortality and morbidity in persons with PAD (63,64). The blood pressure should be reduced to <140/90 mm Hg and to <130/80 mm Hg in patients with diabetes mellitus or chronic renal insufficiency.

Angiotensin-converting enzyme inhibitors reduce cardiovascular morbidity and mortality in patients with PAD regardless of the presence or absence of hypertension (12). The ACC/AHA guidelines recommend treating persons with PAD with angiotensin-converting enzyme inhibitors unless there are contraindications to the use of these drugs to reduce cardiovascular mortality and morbidity (65).

Elderly persons with diabetes mellitus and PAD and no CAD have a 1.5 times higher incidence of new coronary events than elderly nondiabetics with PAD and prior MI (66). The higher the hemoglobin A_{1c} levels in patients with diabetes mellitus and PAD, the higher the prevalence of severe PAD (67). Diabetes mellitus should be treated with the hemoglobin A_{1c} level decreased to less than 7% to decrease the incidence of MI (68). The blood pressure should be reduced to less than 130/80 mm Hg in elderly persons with PAD and diabetes mellitus (69,63,64). Elderly diabetics with PAD should also be treated with statins (70-72) and the serum LDL cholesterol reduced to <70 mg/dL (72).

Increased plasma homocysteine level is a risk factor for PAD (36,32). Reduction of increased plasma homocysteine levels can be achieved by administering a combination of folic acid, vitamin B6, and vitamin B12. However, we do not have double-blind, randomized, placebo-controlled data showing that reduction of increased plasma homocysteine levels will reduce coronary events and slow progression of PAD in elderly persons with PAD.

Hypothyroidism is a risk factor for PAD (37). Elderly persons with clinical or subclinical hypothyroidism should be treated with l-thyroxine to decrease the development of CAD (73). There is no evidence showing that treatment with l-thyroxine will reduce the development of PAD or improve symptoms.

Indications for lower extremity percutaneous transluminal angioplasty or bypass surgery are (1) incapacitating claudication in

persons interfering with work or lifestyle; (2) limb salvage in persons with limb-threatening ischemia as manifested by rest pain, nonhealing ulcers, and/or infection or gangrene; and (3) vasculogenic impotence (74). Percutaneous transluminal angioplasty can be performed if there is a skilled vascular interventionalist and the arterial disease is localized to a vessel segment less than 10 cm in length (74). Compared to percutaneous transluminal angioplasty alone, stenting improves 3-year patency by 26% (75).

Percutaneous balloon angioplasty and/or stenting is indicated for short-segment stenoses, whereas multisegment disease and occlusions are most effectively treated with surgical revascularization (76). In patients presenting with severe limb ischemia caused by infra-inguinal disease and who are suitable for either surgery or angioplasty, by-pass surgery and balloon-angioplasty are associated with similar outcomes in terms of amputation-free survival (77). Patients with intermittent claudication should be considered for revascularization to improve symptoms only in the absence of other disease that would limit exercise improvement such as angina pectoris, heart failure, chronic pulmonary disease, or orthopedic limitations (27).

In patients with critical limb ischemia, when revascularization therapy is not possible, an attempt should be made to avoid amputation with conservative treatment using analgesics, vasodilators and/or anticoagulants. In case of an acute onset of critical limb ischemia, thrombolysis is indicated (21).

Nonrandomized studies have demonstrated that both immediate and long-term survival is higher in patients having revascularization rather than amputation for limb-threatening ischemia (78,79). However, amputation of lower extremities should be performed if tissue loss has progressed beyond the point of salvage, if surgery is too risky, if life expectancy is very low, or if functional limitations diminish the benefit of limb salvage (50).

This is a contemporary approach to establishing an appropriate diagnosis and subsequent treatment of PAD in either ancient or contemporary patients. Treatment of PAD aims to maintain or improve functional status, reduce or eliminate ischemic symptoms, and prevent disease progression. Exercise and aggressive risk factor modification represent the cornerstones of treatment.

Summary

This research analyzes the most likely disease affecting King Asa. The sentence “*Nevertheless in the time of his old age he was diseased in his legs*” indicates that King Asa suffered from a disease

of the legs. Among the many diseases that could have affected the King's legs, gout, degenerative osteoarthritis, and PAD can be considered. Of these possible diagnoses, PAD is the most likely (80).

The diagnosis of PAD requires an extensive evaluation. Although new diagnostic and treatment strategies have developed over time, both ancient and modern patients deserve similar diagnostic investigation and subsequent treatment.

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JACOB'S HEART FAINTS

This section deals with Jacob, the son of Isaac and Rebecca and the third of the Patriarchs of the Jewish people. After Jacob's sons left Egypt, they returned to their father, Jacob and told him that their brother: "*Joseph is still alive, and he is governor over the lands of Egypt. And Jacob's heart fainted, for he believed them not*" (Genesis 45:26). We are told that Jacob fainted, and a transient syncope was responsible for this situation. Among various causes of syncope that could have affected Jacob, the most likely are vasovagal syncope (VVS) or some cardiac arrhythmia.

From the Biblical Story

After the brothers left Egypt, they came back to their father, Jacob. They told him that: "*Joseph is still alive, and he is governor over the lands of Egypt. And Jacob's heart fainted (ויפג לבו) for he believed them not*" (45:26). It was unbelievable news.

How can we understand these two words "ויפג לבו". The Complete Hebrew-English Dictionary explains these words as "his heart failed" (1). The English version of the Bible translates these two Hebrew words as "his heart fainted" (2). We understand from this, that Jacobs' heart failed to work properly and Jacob fainted. What was the condition that afflicted Jacob?

Syncope

Syncope is defined as a temporary interruption of cerebral perfusion with a sudden and transient loss of consciousness and spontaneous recovery (3-6,7).

Since Jacob's heart failed and Jacob fainted, syncope was apparently responsible for this situation.

Causes

The principal groups of causes include (8):

[1] Cardiac causes

- Structural cardiac or cardiopulmonary disease (aortic stenosis, mitral stenosis, pulmonary stenosis, left atrial myxoma, aortic dissection, acute myocardial infarction, cardiac tamponade, pulmonary embolism, obstructive cardiomyopathy)

- Cardiac arrhythmias (tachyarrhythmias, bradyarrhythmias)

- Neurally mediated syncopal syndrome (neurocardiogenic or vasovagal syncope, carotid sinus syncope, and situational syncope)

- Orthostatic (or postural) hypotension

[2] Metabolic causes

- Hypoxia

- Hypoglycaemia

- Hyperventilation

[3] Psychiatric causes

- Somatisation disorders

- Hysteria

- Panic

- Fright

[4] Neurological causes

- Seizure disorders

- Transient ischaemic attacks

- Subclavian steal syndrome

- Normal pressure hydrocephalus

What type of syncope affected Jacob?

Prevalence

Approximately one third of the population experiences syncope at least once during a lifetime (7). Surveys indicate that syncope accounts for approximately 1% of emergency department visits in Europe, although other reports from the United States place this number closer than 3%. The prevalence of syncope in the population varies: in 15% of children before the age of 18 years; 25% of a military population aged 17 to 26 years; in men and women aged 40 to 59 years, 16% and 19%, respectively, and up to 23% in a nursing home population older than 70 years. The highest frequency of syncope occurs in patients with cardiovascular comorbidity and older patients in institutional settings (9).

Orthostatic hypotension affects 5% of the population and 7-17% of patients in acute care settings. It is more common in elderly people and is attributed to an age related decrease in physiological function (reduction in baroreceptor sensitivity) and polypharmacy (including various vasoactive drugs) (10).

The mortality due to syncope varies according to the cause: cardiac causes (arrhythmias or cardiovascular disease) have a 20-30% mortality compared with 5-10% mortality for non-cardiac causes. Although frequent or recurrent episodes can negatively affect quality of life and employability, neurocardiogenic syncope is generally considered a benign condition, as episodes are self-limiting (11).

Mechanism

Stimulation of the cardiac C fibers is implicated in neurocardiogenic syncope (12). An abnormal autonomic response occurs, resulting in vasodilatation and increased vagal tone, with subsequent reduction in cardiac filling and bradycardia, which ultimately leads to syncope. Stimulation of the medullary vasodepressor region of the brain stem may occur owing to activation of various receptors, such as cardiac C fibres (mechanocardiac receptors), cardiopulmonary baroreceptors, cranial nerves, cerebral cortex, and gastrointestinal or genitourinary mechanoreceptors (3, 12,13).

In hypovolaemia and other conditions of reduced preload sympathetic tone is increased, resulting in hypercontractility of the volume depleted ventricle (with increase in myocardial inotropy and chronotropy), with subsequent stimulation of the cardiac C fibres (11,12,14). This results in a combination of parasympathetic enhancement (bradycardia) and decreased sympathetic tone (hypotension), with ensuing syncope (11,14).

Syncope is a syndrome consisting of a relatively short period of temporary and self-limited loss of consciousness caused by transient diminution of blood flow to the brain (most often the result of systemic hypotension) (9).

Syncope may represent a life-threatening condition or a benign process (6,15). Many patients experience neurally mediated (such as neurocardiogenic or vasovagal) syncope. In these patients, cardiovascular control may be stable for an extended period during orthostatic stress, and then there is a sudden decrease in blood pressure and heart rate. Neurocardiogenic or neurally mediated syncope can be associated with painful or emotionally stressful situations such as anxiety or fear, with prolonged standing or specific trigger situations such as micturition, defecation, coughing or sneezing, visceral or carotid sinus stimulation, or with trigeminal or glossopharyngeal neuralgia (7).

Orthostatic hypotension - one of the most frequent causes of syncope - has manifold etiologies comprising various neurological and internal diseases. Orthostatic hypotension usually can be attributed to an impairment of peripheral vasoconstriction or to a reduction of the intravascular volume. Signs and symptoms are often present just after rising from a supine or sitting position. Frequently, blood pressure decreases significantly without an increase in heart rate. Autonomic cardiovascular modulation is often reduced (7).

An initial thorough history and physical examination of the patient will suggest the diagnosis of cardiogenic syncope in at least 50% of patients. A sudden occurrence of syncope, or "drop attack" is most suggestive of an arrhythmic cause for syncope (16).

Orthostatic hypotension is a drop in blood pressure on assuming an upright posture and is due to failure of the autonomic system to compensate for venous pooling in the lower extremities, which results in reduced venous return, decreased cardiac output, and cerebral hypoperfusion (10). Carotid sinus massage is performed to rule out carotid sinus syndrome or hypersensitivity as the cause of syncope. This procedure should be avoided in patients with carotid bruits or a history of cerebrovascular events or transient ischemic attacks, because of the risk of neurological complications (13). Carotid sinus hypersensitivity is defined as "syncope or presyncope resulting from an extreme reflex response to carotid sinus stimulation" (17).

In addition, many medications, including antihypertensives, antidepressants, and digitalis, may be responsible for episodes of fainting (18).

Mechanism of Syncope in Jacob

Jacob's sons described their father as a healthy man. Jacob was not suffering from some cardiovascular, neurological or other disease, so, it is unlikely that the syncope was related to one of these diseases. After many years of mourning for his beloved Joseph, Jacob heard from his sons that Joseph is alive. This unbelievable news acted as a shock that affected Jacob's heart. An immediate response to this positive but stressful news was the inhibition of the sympathetic system and the activation of the vagal system. These mechanisms led to the slowing of the heart rate and a subsequent faint.

The Etiologic Diagnosis

Because of the large differential diagnosis, many diagnostic tests are available for its evaluation. However, a cause of syncope cannot be established in 38% to 47% of patients despite many diagnostic tests (15).

Assessment of the symptoms may yield clues as to the possible cause of the syncope. Syncope after cough, defecation, and micturition suggests situational syncope; syncope associated with throat or facial pain (glossopharyngeal or trigeminal neuralgia) is indicative of neurally mediated syncope with neuralgia; and syncope after pain, fear, or noxious stimuli suggests neurocardiogenic syncope (13). Carotid sinus syncope may occur with rotation or turning of the head or pressure on the carotid sinus (for example, carotid massage, shaving, tight collars or neckwear, or tumor compression) (13).

Patients (n=194) aged 10-92 years were recruited during one year, Jesi Ancona, Italy. The etiological diagnosis included 68.5% cardiovascular causes – 31.9% cardiac, and 36.6% reflex syncope. Non-cardiovascular causes (20.6%) were - 5.2% metabolic, 3.5% neurological, and 2.1% others, while 20.6% were unknown causes. Cardiac syncope has a serious prognosis when it is not treated correctly, and it can even lead to sudden death (19).

Data from case notes and from the autonomic database on 641 patients, at tertiary referral autonomic centers between 1992 and 1998, were analyzed. Syncopal patients with a known or provisional diagnosis of autonomic failure were excluded from analysis. The role of screening tests in establishing or excluding an autonomic cause was assessed. Response to additional autonomic tests such as head-up tilt (HUT) with or without venepuncture, and food challenge and exercise was documented. Some patients underwent further testing if non-autonomic neurological, psychiatric and other disorders were

considered. Autonomic function tests indicated orthostatic hypotension and confirmed chronic autonomic failure in 31 (4.8%) patients. Neurally mediated syncope was diagnosed in 279 (43.5%) patients based on clinical features and autonomic testing. Most had vasovagal syncope (227 [35%]); other causes included carotid sinus hypersensitivity (37 [5.8%]), and a group of 15 (2.3%) were associated with rarer causes such as micturition and swallowing. Miscellaneous cardiovascular causes (systemic hypotension, arrhythmias), or drugs contributed to syncope in 53 (8.3%). Non-autonomic neurological causes included vestibular dysfunction (32, [5%]) and epilepsy (11 [1.7%]). In 56 (8.7%), a psychiatric cause was thought to be contributory. In 179 (27.9%), syncope was of unknown cause. In recurrent syncope and presyncope, when cardiac, neurological, and metabolic causes have been excluded, autonomic investigation can aid management by making, confirming, or excluding various factors or diagnoses (20).

Cardiovascular autonomic nervous system (ANS) function was evaluated extensively in 521 outpatients by careful history, and physical examination included orthostatic blood pressure measurement and standard ECG, and tilt testing. Causes of syncope remained unknown in 29.2% of cases. ANS dysfunction was found in 58.6% of those presenting with either neurally mediated syncope (53.6%) or chronic autonomic failure (5%); 3.8% of the patients suffered from syncope of cardiogenic origin (2.5%) or non-neurogenic hypotension (1.3%), and 8.4% had loss of consciousness of non-syncopeal origin. Loss of consciousness was confirmed as being related to seizures in fewer than 30% of patients initially diagnosed as having epilepsy. Neurally mediated syncope represents the commonest type of syncope. ANS evaluation, including HUT, should be considered as preliminary screening in patients with syncope in the absence of definite heart abnormalities (21).

The medical records of 376 patients with a diagnosis of syncope discharged from the Soroka University Medical Center in 1999 were reviewed. Discharge etiologies were as follows: 26.6% vasovagal, 17.3% cardiac, 4.3% neurological, 0.5% metabolic, 47.3% unexplained, and 4% other; 345 patients were admitted to the internal department, 28 to the intensive cardiac care unit, and 3 to the neurology department. Short-term mortality rates (30 days after discharge) and long-term mortality rates, 1 year after discharge) were 1.9% and 8.8% respectively, and differed according to discharge etiology. Long-term mortality rates were significantly higher in patients discharged with cardiac, neurological and unknown etiologies

(not for vasovagal), compared with the general population of Israel (1 year mortality rate for the age-adjusted [65 years] general population =2.2%). The long-term mortality rate was higher in patients discharged with a cardiac etiology than in those with a non-cardiac etiology (15.4% vs. 7.4%, $p=0.04$). In spite of high costs of inpatients evaluation, associated with more diagnostic tests, longer in-hospital stay and higher mortality rates, nearly half of the patients were discharged without an identified cause (22).

A review of medical records was conducted retrospectively at the Department of Medicine, Aga Khan University Hospital, Karachi. Patients ($n=269$, 75% males, mean age: 57.4 years) aged 16 and above, admitted from January 2000 to December 2005 with the diagnosis of syncope were included. Neurogenic (vasovagal) syncope was the most common cause (47%), followed by cardiogenic syncope (18%) and orthostatic syncope (9%); 24% were discharged undiagnosed. Twenty patients (7.4%) did not have any prodrome. Common prodromal symptoms included dizziness (61%), sweating (25%), palpitations (19%), nausea/vomiting (19%) and visual symptoms (17%). Only breathlessness was significantly associated with cardiogenic syncope ($p=0.002$). Most patients with cardiogenic syncope were aged above 40 (98%, $p<0.001$), had coronary artery disease (72%, $p<0.001$) and abnormal electrocardiogram at presentation (92%, $p<0.001$) (23).

Syncope comprises part of subset of clinical conditions in which loss of consciousness is transient. Other conditions in these groups, which are not syncope and should be clearly distinguished from syncope, include seizure disorders, posttraumatic loss of consciousness, and cataplexy (9,24).

The biblical text states that Jacob was in good health: "... *our father is in good health...*" (43:28). Thus, any of the diseases or the use of medications, which may cause syncope, seems unlikely. After this exclusion, three conditions remain: neurally mediated reflex syncope (e.g., vasovagal faint, or carotid sinus syndrome), orthostatic (postural) syncope, or cardiac arrhythmias. Both carotid sinus syndrome and orthostatic syncope seem unlikely. Thus, cardiac arrhythmia or VVS could have afflicted Jacob. According to Roguin (25), these two conditions are mostly compatible with the biblical description of a medical situation that affected Jacob.

Vasovagal Syncope

The pathophysiology of syncope remains elusive. Lewis introduced the term "vasovagal" implying that both vasodilatation and bradycardia were involved in the response. Susceptible individuals are unable to maintain adaptive neurocardiovascular responses to upright posture for prolonged periods. A complex hemodynamic response develops, with marked hypotension, bradycardia and loss of consciousness. The "empty ventricle theory", first proposed by Sharpey-Schafer, widely accepted for several years, has been challenged and various other aspects of the vasovagal responses have been studied and implicated in contributing to the episodes of unconsciousness. These include baroreflex dysfunction, neuro-endocrine responses, role of respiration and cerebrovascular dysfunction. An episode of syncope represents an episode of unconsciousness. Even a single episode of unconsciousness in the present day lifestyle is a source of distress to patients, warranting a workup and diagnosis. The etiopathogenesis of the simple faint is complex and we may well be dealing with a constellation of responses and a more detailed classification than hitherto imagined (26).

VVS appears to have a benign prognosis (27). Classical (emotional and orthostatic) VVS is not a disease, but rather a manifestation of a non-pathological trait. It is, therefore, reasonable to investigate the possible factors that may explain its origin and evolution. Two processes appear relevant to the investigation of VVS evolution: fear and threat bradycardia in animals, and the vasovagal reflex during hemorrhagic shock in humans and animals. VVS in humans involves physiological mechanisms similar to those found in other vertebrates, and this may indicate common evolutionary roots. The available data suggest that VVS evolved as an advantageous response to inescapable or to stressful and possibly dangerous heart conditions. The inhibition of the sympathetic system, together with activation of the vagal system, characterizes VVS. The consequent slowing of the heart rate induced by VVS may constitute a beneficial break of the cardiac pump, thereby reducing myocardial oxygen consumption. Classical VVS did not evolve recently in the modern human lineage; rather, it should be regarded as a selected response, which probably evolved in the ancient past as a "defense mechanism" of the organism within some ancestral group(s) of vertebrates (26).

VVS is not generally associated with cardiovascular, neurological or other disease, and, therefore, represents an isolated manifestation. Isolated VVS cannot be regarded as a disease for several reasons: spontaneous syncope occurs in about half of

individuals during their lives, and the unidentified neural pathways involved in the vasovagal response are probably present in all healthy humans, with individual differences in susceptibility; VVS is induced during tilt testing in several subjects with no history of syncope; during hemorrhagic shock, the vasovagal reaction can be observed in subjects with no history of syncope; about 20% of astronauts, who are selected on the basis of their great resistance to orthostatic stress, experience syncope or presyncope on landing after a short-duration space flight; to date, no genetic basis of VVS has been demonstrated; subjects with VVS are generally normotensive and, importantly, have normal blood pressure regulation apart from the episodes of syncope; hormonal disorders of a generalized state of autonomic involvement, although frequently investigated, have never been clearly demonstrated (28).

Isolated VVS should be distinguished from those that start in old age and which are often associated with cardiovascular and neurological disorders and other dysautonomic disturbances such as carotid sinus hypersensitivity, post-prandial hypotension, and symptoms of autonomic dysfunction. In these subjects, VVS is an expression of a pathological process, i.e. a disease, mainly related to a generalized involvement of the autonomic nervous system, which is not yet well defined from a nosological point of view (29).

Syncope due to Cardiac Arrhythmia

A long-term subcutaneous monitoring device was implanted in patients with negative ambulatory monitoring, tilt table and electrophysiological testing to establish cardiac rhythm during spontaneous syncope. Sixteen patients aged 57 $\pm$ 19 years with a mean of 8.4 $\pm$ 4.4 previous episodes of syncope underwent device implantation. Syncope was secondary to sinus arrest in 5, atrioventricular block in 2, ventricular tachycardia in 1, supraventricular tachycardia in 1, and no arrhythmia in 6. Unexplained syncope in patients with negative investigations has a broad spectrum of etiologies, the most common of which is bradycardia (30).

There are conflicting opinions on whether postural tachycardia syndrome predisposes to syncope. Autonomic laboratory database of 3700 patients was evaluated. Orthostatic hypotension and postural tachycardia syndrome were defined in standard fashion, except that postural tachycardia syndrome required the presence of orthostatic symptoms and a further increase in heart rate beyond 10 minutes. Syncope was defined as an abrupt decrease in blood pressure and

often, heart rate, requiring termination of the tilt study. Of 810 patients referred for postural tachycardia syndrome, 185 met criteria while another 328 patients had orthostatic hypotension. Of the postural tachycardia syndrome patients, 38% had syncope on HUT, compared with only 22% of those with orthostatic hypotension ($p<0.0001$). In the postural tachycardia group, syncope on HUT was associated with a clinical history of syncope in 90%, whereas absence of syncope on HUT was associated with a clinical history of syncope in 30% ($p<0.0001$). By contrast, syncope on HUT did not bear any relationship to clinical history of syncope in the orthostatic hypotension group (41% vs. 36%; $p=0.49$). Syncope (both tilt table and clinical) occurs far more commonly in patients who have postural tachycardia syndrome than in patients with orthostatic hypotension. These findings suggest that one should be clinically aware of the high risk of syncope in patients with postural tachycardia syndrome. While the low-pressure baroreceptor system, implicated in postural tachycardia syndrome, might confer more sensitivity to syncope than the high pressure system implicated in orthostatic hypotension (31).

Was some type of arrhythmia responsible for syncope in Jacob's case?

Head-up Tilt Test

The HUT has been used to study the hemodynamic and endocrine adaptation to changes in position. There is a substantial agreement that tilt table testing is an effective technique for providing direct diagnostic evidence indicating susceptibility to vasovagal syncope. HUT duration of 30 to 45 min at 60 to 80 degrees has become widely accepted in laboratories for evaluating adult patients. It exhibits a high level of diagnostic specificity of 80% to 100%. Its sensitivity, by contrast, remains 40% to 70%. Pharmacologic provocation during head-up tilt testing elicits susceptibility to vasovagal reactions. HUT is warranted in following conditions: recurrent syncope or single episode of syncope in a high-risk patient, evaluation of patients in whom demonstration of susceptibility to neurally mediated syncope would affect treatment plans, and exercise induced syncope (32).



The evaluation of syncope

HUT is used in the evaluation of patients with syncope to provoke hypotension and/or bradycardia in the laboratory. The most common type of response is provocation of neurally mediated syndrome (VVS). The American College of Cardiology Expert Consensus has proposed indications for tilt testing. The most common indication is recurrent syncope of unexplained cause. In patients with unexplained syncope, positive responses are 50% without the use of isoproterenol and 64% with the use of isoproterenol. Beta-blockers and fludrocortisone plus salt are the most commonly used drugs. Pacemakers have been used, but their role is ill defined (32).

The importance of HUT has been demonstrated in the evaluation of patients with unexplained syncope. Consecutive patients (n=46) aged >65 years (74 +/- 6 years, 56.5% female) with unexplained syncope were studied. In 25 patients (54%), cardiac anomaly was not found. The remaining patients had hypertensive (14 patients) or ischemic (7 patients) heart disease. The protocol included carotid sinus massage in decubitus and after passive orthostatism. HUT was performed with continuous electrocardiographic and blood pressure monitoring, at a 70 degrees tilt for 20 minutes. In the absence of syncope, 500 mcg of nitroglycerin with an additional 20 minutes monitoring was administered. HUT was considered positive when there was reproduction of symptoms associated with bradycardia and/or arterial hypotension (cardioinhibitory, vasodepressor or mixed response). Orthostatic hypotension was defined when systolic blood pressure decreased by >20 mmHg or diastolic blood pressure decreased by >10 mmHg during the first 3 minutes after orthostatism; patients with symptoms associated with a progressive and parallel decrease of systolic and diastolic blood pressures during passive tilt testing were considered to have a dysautonomic profile; patients with symptoms and a gradual decrease of blood pressure after nitroglycerin were considered to have an exaggerated response to nitrates. Symptoms were reproduced in 34 patients (73.9%): 19.6% during passive tilting (neurocardiogenic vasodepressor response - 3 patients,

carotid sinus hypersensitivity - 1 patient, orthostatic hypotension -1 patient, dysautonomic profile - 4 patients) and 54.3% after nitroglycerin (neurocardiogenic vasodepressor response - 12 patients, mixed - 5 patients, cardioinhibitory - 2 patients and exaggerated response to nitrates - 6 patients). Syncope was neurocardiogenic in 47.8% (13.6% in passive test, and 86.4% after nitroglycerin). There was an exaggerated response to nitrates in 16.2% of the patients given nitroglycerin. There were no complications during the test. HUT in the elderly with syncope of unknown origin helps differential diagnosis in etiological study of the syncope and when potentiated by nitroglycerin, it produces a significant increase in positive responses, and identifies a considerable number of patients with an exaggerated response to nitrates (34).

HUT is used primarily for the investigation of orthostatic symptoms. Although this test is frequently the gold standard for the evaluation of neurocardiogenic syncope, and dysautonomia or postural orthostatic tachycardia syndrome, there is a debate over its diagnostic value and method. Despite its limitations, the HUT test is useful in patients with a variety of clinical manifestations induced by orthostatism. It is most useful in documenting objective measures of orthostatic hypertension that cannot be obtained in a clinical setting (35).

Some patients with neurally mediated reflex syncope may be misdiagnosed as epilepsy because myoclonic jerky movements are observed during syncope. The medical records of 1,383 consecutive patients with a positive HUT were retrospectively reviewed, and 226 patients were included in this study. Of 226 patients, 13 (5.8%) showed seizure-like activities, 5 (2.21%) had multifocal myoclonic jerky movements, 5 (2.2%) demonstrated focal seizure-like activity involving one extremity, and 3 (1.3%) showed upward deviation of eye ball. Comparison of patients with and without seizure-like activities revealed insignificant differences in terms of clinical variables and hemodynamic parameters during HUT. Seizure-like activities occurred occasionally during HUT-induced syncope in patients with neurally mediated reflex syncope. The seizure-like activities during HUT might not be related to the severity of the syncopal episodes or hemodynamic changes during HUT (36).

Outcome of Syncope

Syncope is a common presenting complaint to the emergency department (ED). Syncope causes are many and varied, some of which are potentially life threatening. While the cause of syncope can be identified in many patients, and life-threatening conditions subsequently treated, a risk stratification approach should be taken for those in whom a cause is not identified. Identifying a cardiac cause for syncope is a poor prognostic indicator. Patients with unexplained syncope who have significant cardiac disease should therefore be investigated thoroughly to determine the nature of the underlying heart disease and the cause of syncope, although there is little evidence that this improves their dismal prognosis (37).

Its assessment is difficult. Some serious causes of syncope are transient and patients with a potentially life threatening condition may appear well by the time they reach the ED. Accurate history taking is vital and is often diagnostic whilst identification of a cardiac cause is associated with an increased mortality. This is related to underlying cardiac disease; patients presenting with syncope, who have significant cardiac disease, should be investigated thoroughly to determine the nature of the underlying heart disease and the cause of syncope. Patients (n=30%) with cardiac syncope die within one year of presentation. It is likely that serious outcome in syncope although significant, is not quite as common as previously thought (37).

This study was conducted as a feasibility pilot for the Risk stratification Of Syncope in the Emergency department (ROSE) study. The secondary aim was to compare the performance of existing ED guidelines with existing clinical decision rules (Osservatorio Epidemiologico sulla Sincope nel Lazio (OESIL) Score and San Francisco Syncope Rule; SFSR) at predicting short-term (1 week and 1 month) and medium-term (3 months) serious outcomes for patients with syncope presenting to the ED. All patients presenting with syncope, aged ≥ 16 years between 7 November 2005 and 7 February 2006, were prospectively enrolled; 99 patients were recruited over a 3-month period; 44 patients were admitted and 55 discharged from the ED. 11 patients had a serious outcome: 8 by 7 days and a further 3 by 3 months. Five patients died by 3 months and six others had an alternative serious outcome. Percentages of serious outcomes were 0%, 2.9%, 8.0%, 22.7% and 37.5% for OESIL scores of 0, 1, 2, 3 and 4 respectively. Patients (n=40) had none of the 5 SFSR high-risk factors (0 serious outcomes = 0%) and 59 patients had an SFSR high-risk factor (11 serious outcomes = 18.6%). The risk of serious

outcome at 7 days, 1 month and 3 months was 8.1%, 8.1% and 11.1%, respectively (38).

Management

How these patients should be managed? The mainstay of management is education of the patient to avoid situations that predispose to syncope, with anxiety management, coping skills, and reassurance of the patient and others that this is a benign condition (8). The diagnosis of the cause of syncope is based on a careful clinical history obtained from the patients or from persons who attended one of the episodes of loss of consciousness. It should also include a careful clinical examination including an electrocardiogram and, most often, an echocardiogram. The selection of further investigations should be guided by the existence or absence of underlying cardiac disease (39).

If there is no evidence of neurological, metabolic, or obvious cardiac (for example, aortic stenosis) cause for syncope, a 24-hour ambulatory ECG should be performed. If the initial 24-hour ambulatory ECG is negative and if the patient has had multiple episodes of syncope, multiple days of recording will be necessary before typical symptoms occur that correlate with arrhythmias (16). Echocardiography and exercise stress testing are not routinely indicated in the evaluation of syncope unless the clinical history and examination suggest valvular or coronary disease (16,40).

The optimal management of patients presenting with syncope depends on the etiology of symptoms. The first step is to differentiate patients without cardiac diseases from others. The clinical history and the examination are the most useful, and in most half the cases, suggest a presumptive diagnosis. In patients without heart disease, tilt test and autonomic nervous system testing are the investigations with the greatest yield. In patients with heart disease, hemodynamic and electrophysiologic studies are frequently needed. Neurological investigations are rarely useful but only if syncope is associated with focal neurological manifestations (41).

Drug-induced autonomic failure is probably the most frequent cause of orthostatic hypotension. The principal treatment strategy is elimination of the offending agents, mainly diuretics and vasodilators. Alcohol is also commonly associated with orthostatic intolerance. Additional treatment principles, used alone or in combination, are appropriate for consideration on an individual patient basis: chronic expansion of intravascular volume by encouraging a higher than normal salt intake and fluid intake of 2–2.5 liters per day (42,43).

For the treatment of syncope, one should first consider non-pharmacological options. Patients with orthostatic hypotension should avoid rapid changes of the body position from supine to standing, as well as high room temperature or other situations inducing peripheral vasodilatation. An increased intake of sodium and fluids, mild physical exercise or so-called postural counter-maneuvers can improve orthostatic tolerance. Among the drugs recommended for pharmacologic treatment are mineralocorticoids (e.g. fludrocortisone), vasoconstrictor agents (e.g. ephedrine, midodrine), adenosine receptor blockers (theophylline) and beta₂-blockers (propranolol), anticholinergic agents, e.g. scopolamine or disopyramide, and negative cardiac inotropes, e.g. beta₁-adrenergic blockers or disopyramide. Serotonin reuptake inhibitors (e.g. fluoxetine, sertraline), alpha₂-adrenergic agonists (clonidine), central nervous system stimulants such as methylphenidate or phentermine are beneficial in specific cases. Cardiac pacemakers often seem to be recommended without adequate indication. The antidiuretic, V2-receptor specific, vasopressin analogue desmopressin increases the intravascular volume. Erythropoietin improves anemia and red blood cell decrease and augments blood pressure and cerebral oxygenation. In postprandial hypotension, octreotide, a somatostatin analogue, prostaglandin inhibitors such as indomethacin or ibuprofen, as well as metoclopramide or two cups of coffee per day might be beneficial (7).

In some patients, implantable ECG loop recorder is useful. In most patients, correlation of symptoms and results of investigations lead to appropriate therapeutic strategy (41).

To sum up: when Joseph's brothers returned from Egypt to their father Jacob, they told him "*Joseph is still alive, and he is governor over the lands of Egypt. And Jacob's heart fainted*, (ויפג לבו) *for he believed them not*" (45:26). This biblical description is compatible with syncope. Among various causes of syncope that most likely afflicted Jacob such as (1) neurally mediated reflex syncope, e.g. vasovagal faint, carotid sinus syndrome; (2) orthostatic (postural) syncope; (3) cardiac arrhythmias; (4) structural cardiac and pulmonary causes; (5) and cerebrovascular disorders (rare), the most likely are VVS syncope or syncope due to some cardiac arrhythmia. A contemporary patient similarly to ancient patient may suffer from syncope, and they deserve an appropriate evaluation and treatment.

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WHAT WAS THE HEART DISEASE THAT AFFLICTED A BIBLICAL FIGURE?

Introduction

Patients have died from various heart diseases for thousands of years. Who died from heart disease in biblical times? What was a most likely cause of this death? What was the characteristic of such patient? What was the most likely diagnosis? Were any complications present?

This article aims to answer these questions by evaluating the biblical description of a man who died from some type of heart disease. Evaluation of an ancient patient who was afflicted by some heart disease may widen the horizons of knowledge of contemporary physicians and help them to treat their patients more successfully.

The Biblical Story

Nabal was a very rich man, while his wife Abigail was a wise and beautiful woman. But he was a very hostile personality, *"hardhearted and evil in his doing"* (I Samuel 25:3). At that time, David's men saved Nabal's shepherds from robbers. On the feast day of shearing the sheep, David sent ten men to Nabal with greetings in order to receive the payment for their service: *"A hearty greeting! Peace be both to thee, and peace to thy house, and peace to all that thou hast"*. Nevertheless, Nabal scorned David's greetings: *"Who is David? There are many servants nowadays that break away every man from his master. Shall I then take my bread, and my water, and*

the fresh meat that I have killed for my shearers, and give it to men from I know not where?” (25:10,11).

David was furious and was eager for revenge. On hearing that, Abigail decided to end the conflict peacefully. She secretly took a present to David “...two hundred loaves, and two bottles of wine, and five sheep ready prepared, and five measures of parched corn, and a hundred clusters of raisins, and two hundred cakes of figs, and laid them on asses” (25:18). Abigail’s meeting with David ended in a peaceful agreement to spare Nabal and his household.



Sir Peter Paul Rubens
The Meeting of David and Abigail, c. 1630.

After this meeting, Abigail came to Nabal, who was heavily drunk. Only in the morning when he awoke did she tell him about David “...when the wine was gone out from Nabal, and his wife had told these things, that his heart died within him, and he became as a stone. And it came to pass about ten days after....he died” (25:37,38). So Nabal died, and after his death, David took Abigail to be his wife.

The Disease that Led to Nabal’s Death

Since Nabal’s past medical history was unremarkable, it can be concluded that he was a middle-aged healthy man. After his wife told him about her meeting with David, Nabal fell into a great rage, which led to severe mental distress at his wife’s betrayal. After a period of very unpleasant psychological distress Nabal died: “...his heart died”....and “...he died.” These biblical words indicate an acute myocardial infarction (AMI).

Psychological Disturbances

Psychosocial factors contribute significantly to the pathogenesis and expression of coronary artery disease (CAD). This evidence is composed largely of data relating CAD risk to 5 specific psychosocial domains: (1) depression, (2) anxiety, (3) personality factors and character traits, (4) social isolation, and (5) chronic life stress. Pathophysiological mechanisms underlying the relationship between these entities and CAD can be divided into behavioral mechanisms, whereby psychosocial conditions contribute to a higher frequency of adverse health behaviors, such as poor diet and smoking, and direct pathophysiological mechanisms, such as neuroendocrine and platelet activation. Specific suggestions to improve the assessment of behavioral interventions include more complete delineation of the physiological mechanisms by which such interventions might work; increased use of new, more convenient "alternative" end points for behavioral intervention trials; development of specifically targeted behavioral interventions (based on profiling of patient factors); and evaluation of previously developed models of predicting behavioral change. The importance of maximizing the efficacy of behavioral interventions is underscored by the recognition that psychosocial stresses tend to cluster together. When they do so, the resultant risk for cardiac events is often substantially elevated, equaling that associated with previously established risk factors for CAD, such as hypertension and hypercholesterolemia (1).

Conventional risk factors (abnormal lipids, hypertension, etc.) are independent predictors of atherosclerosis and cardiovascular events; however, these factors are non-specific since in about half patients AMI paradoxically develops at low cardiovascular risk. Some psychosocial factors are independent predictors of atherosclerosis and cardiovascular events, as well. Psychosocial factors that promote atherosclerosis include two general categories: chronic stressors, such as social isolation/low social support, work stress (subordination without job control) and emotional factors, including affective disorders such as depression, severe anxiety, hostility and anger. The emotional factors, such as the chronic stressors, activate the biological mechanisms of chronic stress: increase activity of the hypothalamic-pituitary-adrenal axis, sympathetic system, and inflammation processes, which have atherogenic effects and an increase in blood coagulation (2).

Increased knowledge concerning the triggering of acute cardiovascular diseases has yielded a change in philosophical approach to this field. The term of acute risk factors could be used for

the activities and events that suddenly and transiently increase the risk of acute cardiac diseases. External triggers, such as heavy physical activity, emotional stress, eating, cold or heat exposure, coffee or alcohol consumption, cocaine or marijuana use and sexual intercourse are the most important acute risk factors. It is likely that the morning hours are an endogenous, external triggering independent acute risk factor related to physiological sympathetic arousal. The features of triggering have been best described for an AMI whose moment of onset appears to be the result of a dynamic interaction between an endogenous response to acute risk factors and patient vulnerability. A highly individual approach tailored both to protect against acute risk factors and to reduce patient vulnerability could provide a more complete protection from AMI and other coronary incidents. Lifestyle modifications, regular physical activity and adequate drug regimens may at least prove to defer the occurrence of coronary thrombosis, thereby providing time for the development of collateral vessels, plaque stabilization or invasive/surgical treatment (3).

Psychological stress significantly influences the pathogenesis of sudden cardiac death, expressing a situation of imbalance, derived from a real or perceived disparity between environmental demands and the individual's ability to cope with these demands. A situation of psychological stress may include different components: personality factors and character traits, anxiety and depression, social isolation and acute or chronic adverse life events. In particular, a sudden extremely hard event, such as an earthquake or a war strike, can significantly increase the incidence of sudden death. Nevertheless, each one of these factors, if not present, can balance a partially unfavorable situation; there is a multifactorial situation where almost all elements are present and in which the relative influence of each one varies according to the individual examined. Sudden death occurs when a transient disruption (such as acute myocardial ischemia, platelet activation or neuroendocrine variations), occurring in a patient with a diseased myocardium (such as one with a post-necrotic scar or hypertrophy), triggers a malignant arrhythmia. Psychological stress acts at both levels: by means of a "chronic" action, it contributes to create the myocardial background, while by means of an acute action it can create the transient trigger precipitating sudden death. In the chronic action, two possible mechanisms can be detected: the first is a direct interaction, which contributes to cause a hypertension status or to exacerbate coronary atherosclerosis consequent to endothelial dysfunction; the second one acts through adverse health behaviors, such as a poor diet, alcohol consumption, or smoking (4).

Three sets of conditions contribute to the occurrence of arrhythmias: myocardial electrical instability, most often due to CAD; an acute triggering event, frequently related to mental stress; and a chronic, pervasive, and intense psychological state, often including depression and hopelessness. The autonomic nervous system plays an important role in the occurrence of cardiac arrhythmias and it is well documented that mood alterations as mental stress and depression influence cardiac autonomic balance. Patients with the greatest changes in cardiac neural regulation with decreased parasympathetic tone coupled with increased sympathetic activity are at the greatest risk for developing fatal ventricular arrhythmias. These patients have reduced heart rate variability, increased QT-dispersion and decreased baroreceptor sensitivity (5).

Anger

Anger refers to an unpleasant emotion, ranging in intensity from irritation to rage, usually in response to perceived mistreatment and provocation (6). In certain cases, emotionally stressful events and, more specifically, episodes of anger immediately precede and trigger the onset of AMI (7-12). The risk of AMI was elevated during the first 2 hours after an outburst of anger, with the most frequent causes of anger being arguments with family members, conflicts at work, and legal problems (12). Moreover, a significant correlation was found between the amount of anger and coronary artery vasoconstriction in previously narrowed arteries, while, in non-narrowed arterial segments there was insignificant relationship between anger and vasomotion (13). Therefore, anger evokes physiological responses that are potentially life threatening in the context of patients afflicted by CAD (14). In addition, anger and other negative emotions can trigger potentially lethal ventricular arrhythmias (15). Intense or unusual emotional states, often involving anger, are associated with arrhythmia onset in approximately 20% of the cases (16).

Patients with implantable cardioverter-defibrillators (ICDs) were given diaries to record levels of defined mood states and physical activity, using a 5-point intensity scale, during 2 periods preceding spontaneously occurring ICD shocks (0 to 15 minutes and 15 minutes to 2 hours) and during control periods 1 week later. ICD-stored electrograms confirmed the rhythm at the time of shock. Ventricular arrhythmias (n=107) requiring shock were reported by 42 patients (33 men; mean age, 65 years; 78% had CAD) between August 1996 and September 1999. In the 15 minutes preceding shock, an anger level ≥ 3 preceded 15% of events compared with 3% of control periods

($p < 0.04$; odds ratio 1.83; 95% CI 1.04-3.16) Other mood states (anxiety, worry, sadness, happiness, challenge, feeling in control, or interest) did not differ significantly. Patients were more physically active preceding shock than in control periods. Anger and physical activity were independently associated with the preshock period. Therefore, anger and physical activity can trigger ventricular arrhythmias in patients with ICDs (15).

We see that Avigail's deceitful behavior led Nabal to extreme anger, which played a role in triggering vasoconstriction of coronary artery(s), and subsequently to AMI most likely accompanied by lethal arrhythmia. Since Nabal's past medical history was unremarkable, it seems unlikely that he was previously afflicted by a significant CAD.

Hostility

High levels of hostility associate with increased levels of arteriographically documented atherosclerosis (17). Hostility and anger precipitate myocardial ischemia, infarction, and life-threatening arrhythmias (18). On the other hand, no positive association was found between hostility and CAD (19)

Nabal was a very hostile man "*hardhearted and evil in his doing*". Therefore, it seems very likely that an extremely hostile personality accompanied by an outburst of anger acted as a trigger in the development of AMI.

Emotional Stress

Psychological stress is associated with increases in cardiac events (12). Acute mental stress triggers transient myocardial ischemia, AMI, and sudden cardiac death (20-22) and therefore can be regarded as a risk factor for AMI (9,23). In approximately 4% to 18% of cases of AMI, emotional stress occurs immediately before the onset of symptoms (8). Feelings of tension, frustration, and sadness are associated with a 2-fold increased risk of myocardial ischemia in the next hour (24). Exceptionally heavy physical work, a violent quarrel at work or at home, and unusual mental stress are the three most frequent triggers reported immediately before or within the 24 hours preceding the onset of pain (9). In addition, excess cardiac mortality attributed to acute psychological stress has been reported following earthquakes and other natural disasters. During the five days after 1981 Athens earthquake, the incidence of cardiac deaths nearly doubled (25). In addition, a 22% increase in CAD deaths was observed following

blizzards in Massachusetts (26). Extreme emotional upset has also been associated with the development of AMI and subsequent cardiogenic shock (27).

Acute coronary syndrome (ACS) is an important entity of the cardiovascular morbidity and mortality. Everyday circumstances, which significantly increase the risk of the onset of ACS during the period of several hours, have been termed acute risk factors. This term includes morning hours and various possible external triggers such as heavy physical exertion, emotional stress, sexual activity, meteorological stress, and large meals. Possible external triggers precede approximately a half of the onsets of ACS, more frequently in men. Conversely, emotional stress plays a more important role in the triggering of ACS in women. Circadian rhythm represents an independent natural pattern of endogenous functions that is responsible for the morning excess in the incidence of ACS. A beneficial effect of regular physical activity and use of beta-blocker, calcium antagonists, nitrates and aspirin in the prevention of triggering ACS by acute risk factors has been described. Identification of further acute risk factors and better understanding of internal triggering mechanisms may improve the prevention of ACS and other cardiovascular incidents (3).

Episodes of acute emotional stress have significant adverse effects on the heart, producing left ventricular (LV) contractile dysfunction, myocardial ischemia, or disturbances of cardiac rhythm. Although these abnormalities are often only transient, their consequences can be gravely damaging and sometimes fatal. Despite the many descriptions of catastrophic cardiovascular events in the setting of acute emotional stress, the anatomical substrate and physiological pathways by which emotional stress triggers cardiovascular events are characterized, and aided by the advent of functional neuroimaging. Asymmetric brain activity is particularly important in making the heart more susceptible to ventricular arrhythmias. Lateralization of cerebral activity during emotional stress may stimulate the heart asymmetrically and produce areas of inhomogeneous repolarization that create electrical instability and facilitate the development of cardiac arrhythmias. Patients with CAD who survive an episode of sudden cardiac death in the setting of acute emotional stress should receive a beta-blocker. Nonpharmacological approaches to manage emotional stress in patients with and without CAD, including social support, relaxation therapy, yoga, meditation, controlled slow breathing, and biofeedback, are also appropriate to consider and merit additional investigation in randomized trials (28).

The forensic pathologist based on autopsy findings and toxicological analysis determines the cause of death in cases of sudden and unexpected death. In cases where no acute pathology or lethal injury are detected, and the circumstances surrounding the death are highly stressful or emotional, it is possible to attribute the inciting event as contributory to the death by causing cardiac dysrhythmia. Ten cases of sudden death related to stressful events were examined over a period of 2 years in the National Center of Forensic Pathology, Israel, all of which lacked findings of acute disease or serious injury. All suffered from variable degrees of chronic cardiac pathology, which probably contributed to dysrhythmia and death. Since the seventies of the previous century, forensic pathologists are entitled to determine mode of death as homicide or manslaughter, due to an acute cardiac event, which cannot be diagnosed in an autopsy and which, was a direct result of the circumstances prior to death. Later, these conditions were adjusted to include cases where no morphological findings were detected or when death is delayed by medical treatment. The physiological mechanisms responsible included increased catecholamine levels, as well as blood pressure and platelet aggregation and a delay in cardiac and vascular recovery from stress. Victims of sudden death caused by emotional stress, during a criminal act, usually have a background of chronic cardiac disease and/or are more susceptible physiologically to suffer an amplified reaction to stress. Legally, their deaths are considered as homicide or manslaughter (29).

It can be therefore concluded that in Nabal's case acute and severe mental stress, caused by Avigail's deceitful behavior, played a role in triggering the development of AMI most likely associated with some fatal arrhythmia.

Depression

CAD and depression are both highly prevalent diseases. Both cause a significant decrease in quality of life for the patient and impose a significant economic burden on society. There are several factors that link depression with the development of CAD and with a worse outcome in patients with established CAD: worse adherence to prescribed medication and life style modifications in depressive patients, as well as higher rates in abnormal platelet function, endothelial dysfunction and lowered heart rate variability. Depression per se is an independent risk factor for cardiac events in a patient population without known CAD and in patients with established

diagnosis of CAD, particularly after AMI. Treatment of depression improves patients' quality of life. However, it did not improve cardiovascular prognosis in depressed patients even though there is open discussion about the trend to better outcome in treated patients. Selective serotonin reuptake inhibitors seem to be preferable to tricyclic antidepressants for treatment of depressive patients with comorbid CAD because of their good tolerability and absence of significant cardiovascular side effects. Hypericum perforatum (St. John's wort), an increasingly used herbal antidepressant drug should be used with caution due to severe and possibly dangerous interaction with cardioactive drugs (30).

CAD, congestive heart failure (CHF) and depression significantly affect quality of life and represent a major economic burden for society. Depression is an independent risk factor for the onset and progression of CAD and CHF. Patients with depression are more likely to develop CAD, and they are more likely to die or to have a recurrence after AMI. CHF is frequently associated with depression, and this combination carries an increased risk of complications and mortality. There is strong evidence that depressive symptoms are a negative prognostic factor in patients with CHF and CAD (31).

Depression and CAD are related in several ways: (1) High levels of depressive symptoms in male and female patients are associated with an increased risk of AMI and a higher mortality following an acute cardiac event. Furthermore, patients developing depression after AMI have more complications, including cardiac arrhythmias (2). In patients with a chronic CAD, depression also results in a worse cardiac functional status with more frequent and severe chest pain, more physical limitation, less treatment satisfaction and a lower perceived quality of life. Non-compliance with drug therapy is also more prevalent in depressed cardiac patients (32).

Did Nabal suffer from depression? There are insufficient data for a diagnosis of this mental disorder.

Mechanism of Injury

Various components of psychological factors are responsible for the fact that life events or certain psychosocial factors can lead to a stress reaction. Psychosocial risk factors have direct pathophysiologic effects and influence life style and behavior. Stress is a reaction of the organism to a stressor with various negative health consequences

including the development of arteriosclerosis. Stress may cause myocardial ischemia, rhythm disturbances, and platelet stimulation, increased blood viscosity through hemoconcentration as well as endothelial dysfunction and coronary vasoconstriction in the presence of arteriosclerosis of the coronary arteries. Psychosocial risk factors may also lead to the development and to clinical consequences of arteriosclerosis (33).

Psychosocial factors like depression, anxiety, social isolation, stress and some types of personality, contribute significantly to the pathogenesis and expression of CAD. Pathophysiological mechanisms in this interrelation can be divided between psychosocial factors conditions that contribute to a higher frequency of adverse health behaviors (e.g. smoking) or directly increasing the platelets and neuroendocrine activity. Acute stress triggers myocardial ischemia, promotes arrhythmogenesis, stimulates platelet function, and increases blood viscosity through hemoconcentration. The foregoing effects are in part a result of the endothelial dysfunction and injury induced by acute stress. Psychosocial stresses tend to cluster together and when these take place, the resultant risk for cardiac events is often substantially elevated, equaling an association with previously established risk factors for CAD, such as hypertension and hypercholesterolemia (34).

The influence of anxiety and depression on cardiovascular disease is known in terms of chronic cardiovascular and ischemic states. It happens mostly through diminished cardiac vagal modulation and greater procoagulant activity in depressed patients. From the clinician's point of view, there are an increasing awareness concerning external circumstances and external triggering in sudden cardiac death, caused by malignant ventricular arrhythmias or AMI. A number of external, i.e. exogenous factors in everyday life can be implicated in triggering such events. Including heavy physical activity, emotional stressors, meteorologic variables, air pollutants, chronobiologic phenomena, food ingestions, coffee, alcohol etc., emotional stress seems to be among the most often triggers of sudden event. There is a wide range of everyday situations and activities that involves activation of sympathetic nervous system, and emotional stressors acting as trigger caused by sympathetic arousal. Acute emotional arousal can trigger life-threatening arrhythmias. Acute episode of emotional distress (and notably anger-like stress) may trigger ventricular ectopic beats and tachycardia, as well as AMI. These sudden phenomena can be the result of the endogenous response to external risk of acute emotional stress. Sudden death without

antecedent angina, suggesting cardiac arrhythmia, as well as deaths preceded by chest pain, suggesting coronary occlusion have led to estimates that between 20% and 40% of sudden cardiac deaths are precipitated by acute emotional stressors (35).

Chronic psychosocial stress can lead via a mechanism, involving excessive sympathetic nervous system activation, to exacerbation of coronary artery atherosclerosis as well as to transient endothelial dysfunction and even necrosis. Psychosocial stress induces ovarian dysfunction, hypercortisolemia, and excessive adrenergic activation in premenopausal females, leading to accelerated atherosclerosis. Acute stress triggers myocardial ischemia, promotes arrhythmogenesis, stimulates platelet function, and increases blood viscosity through hemoconcentration. In the presence of underlying atherosclerosis (e.g., in CAD patients), acute stress also causes coronary vasoconstriction. The foregoing effects result, at least in part, from the endothelial dysfunction and injury induced by acute stress. Hyperresponsivity of the sympathetic nervous system, manifested by exaggerated heart rate and blood pressure responses to psychological stimuli, is an intrinsic characteristic among some individuals. Current data link sympathetic nervous system hyperresponsivity to accelerated development of carotid atherosclerosis in human subjects and to exacerbated coronary and carotid atherosclerosis in monkeys (1).

Anger reduces the LV ejection fraction in patients with 1-vessel CHD (36). Psychosocial stress appears to adversely affect autonomic and hormonal homeostasis, resulting in metabolic abnormalities, inflammation, insulin resistance, and endothelial dysfunction (37). Mental stress increases heart rate, blood pressure, myocardial oxygen demands, coronary flow, coronary vascular resistance and vasoconstriction, and may promote platelet aggregation and coagulation, opposes fibrinolytic factors, and increases plaque rupture (38-44,45). These effects are mediated at least in part by catecholamine secretion (38). Increases in the plasma concentrations of the platelet-secreted proteins platelet factor 4, β -thromboglobulin, epinephrine and norepinephrine are associated with an emotional stress (41). Adrenergic activity, causing ischemia signals of cardiac pain to pain centers via sympathetic efferent pathways, tend to induce arrhythmogenic and necrotizing ischemic actions on the cardiovascular system (42). Some individuals have a sympathetic nervous system hyper-responsibility, manifesting as exaggerated heart rate and blood pressure responses, which result in accelerated atherosclerosis (45).

Anger alters electrophysiological properties of the myocardium, including T-wave alternans (TWA), a measure of heterogeneity of repolarization, suggesting one mechanistic link between emotion and arrhythmia. Anger and other strong emotions can trigger polymorphic, potentially life-threatening ventricular arrhythmias in vulnerable patients. Through autonomic changes, including increased sympathetic activity and vagal withdrawal, anger leads to increases in heterogeneity of repolarization as measured by TWA, known to be associated with arrhythmogenesis, as well as increasing inducibility of arrhythmia (46).

To investigate the role of vigorous physical exertion and anger as triggers of ACS and to identify the clinical and sociodemographic correlates of triggering a prospective observational clinical cohort study was carried out at four coronary care units in the London area. Patients included 295 men and women with electrocardiographically and biochemically verified ACS. Physical exertion in the 1 hour and anger in the 2 hours before symptom onset were assessed with structured interviews. Control periods were the equivalent hours one day earlier and usual rates over the past six months. Physical exertion was reported by 10% and anger by 17.4% of patients in the hazard period. The risk of ACS onset after physical exertion compared with light or no activity was 3.50 (95% confidence interval [CI] 1.37-10.6). The risk of onset with anger was 2.06 (95% CI 1.12-3.92). Physical exertion during the hazard period was related to an absence of premonitory symptoms, presentation with an ST elevation myocardial infarction (STEMI), low socioeconomic deprivation and higher future cardiovascular risk. Anger during the hazard period was more common in younger, socioeconomically deprived patients who presented with a STEMI. Triggers were relevant across the spectrum of ACS. Thus, the distinct clinical and sociodemographic factors associated with physical exertion and anger suggested that different pathophysiological processes were involved (47).

Sixty-two patients with ICDs underwent ambulatory electrocardiography during a mental stress protocol, 3 months after the ICD was implanted. TWA was analyzed using time-domain methods. After a ≥ 1 year of follow-up, ICD stored data was reviewed to determine incidence of ICD-terminated ventricular tachycardia/ventricular fibrillation. Patients with ICD-terminated arrhythmias during follow-up (n=10) had higher TWA induced by anger, 13.2 microV (interquartile range [IQR] 9.3 to 16 microV), compared with those patients without future ventricular arrhythmias, 9.3 microV (IQR 7.5 to 11.5 microV, $p < 0.01$). Patients in the highest

quartile of anger-induced TWA (>11.9 microV, $n=15$) were more likely to experience arrhythmias by 1 year than those in the lower quartiles (33% vs. 4%) and during extended follow-up (40% vs. 9%, $p<0.01$ for both). In multivariable regression controlling for ejection fraction, prior clinical arrhythmia, and wide QRS, anger-induced TWA remained a significant predictor of arrhythmia, with likelihood in the top quartile 10.8 times than that of other patients (95% CI 1.6-113, $p<0.05$). Thus, anger-induced TWA predicts future ventricular arrhythmias in patients with ICDs, suggesting that emotion-induced repolarization instability may be one mechanism linking stress and sudden death (48).

TWA responses to mental stress (anger recall and mental arithmetic) and bicycle ergometry were evaluated in patients with implantable ICDs and documented CAD ($n=23$, age 62.1 ± 12.3 years) and controls ($n=17$, age 54.2 ± 12.1 years). TWA was assessed from digitized ECGs by modified moving average analysis. Dual-isotope single photon emission computed tomography was used to assess myocardial ischemia. TWA increased during mental stress and exercise ($p<0.001$), and TWA responses were higher in ICD patients than in controls (arithmetic $\Delta=8.9\pm 1.4$ versus 4.3 ± 2.2 microV, $p=0.043$; exercise $\Delta=21.4\pm 2.8$ versus 13.8 ± 3.2 microV, $p=0.038$). TWA increases with mental stress, occurring at substantially lower heart rates (anger recall $\Delta=9.7\pm 7.7$ bpm, arithmetic $\Delta=14.3\pm 13.3$ bpm) versus exercise ($\Delta=53.7\pm 22.7$ bpm; $p<0.001$). After adjustment for heart rate increases, mental stress and exercise provoked increased TWA in ICD patients ($p<0.05$), but not in controls ($p>0.2$). Ejection fraction and stress-induced myocardial ischemia were not associated with TWA. Mental stress can induce cardiac electrical instability, as assessed via TWA, among patients with arrhythmic vulnerability and occurs at lower heart rates than with exercise. Pathophysiological mechanisms of mental stress-induced arrhythmias may therefore involve central and autonomic nervous system pathways that differ from exercise-induced arrhythmias (49).

At the time of shock, patients with implantable ICDs recorded levels of defined mood states preceding the shock in a diary. Stored intracardiac electrograms (EGMs) were retrieved and analyzed in relation to corresponding mood states. The EGMs from 56 appropriate shocks in 24 patients (18 male, mean age 66 years, 74% with CAD) were reviewed and analyzed for morphology, mechanism of initiation (sudden onset vs. premature ventricular contraction [PVC]), pause dependence, and other characteristics. Polymorphic ventricular

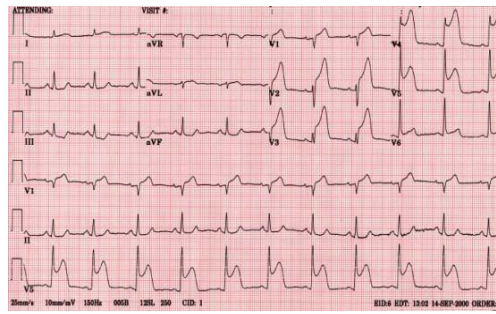
tachycardia (VT) was more common in anger-triggered events, occurring in three (37.5%) of eight anger-triggered events compared with five (10.4%) of 48 of non-anger-triggered events ($p<0.05$). Anger-triggered events were more likely to have PVC initiation, occurring in eight (100%) of eight, compared with 30 (68%) of 44 of non-anger-triggered events ($p<0.05$). More anger-triggered events were pause dependent: five (62.5%) of eight versus seven (15%) of 37 non-anger-related events ($p<0.01$). No difference in response to initial therapy was observed in anger-triggered arrhythmias. Ventricular arrhythmias occurring in the setting of anger were more likely pause dependent and polymorphic. Therefore, in predisposed populations anger may create an arrhythmogenic substrate susceptible to more disorganized rhythms, a possible mechanism linking emotions to sudden death (50).

What was the mechanism responsible for the development of AMI in Nabal's case?

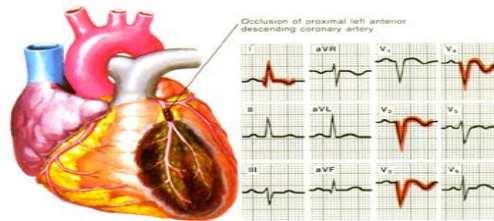
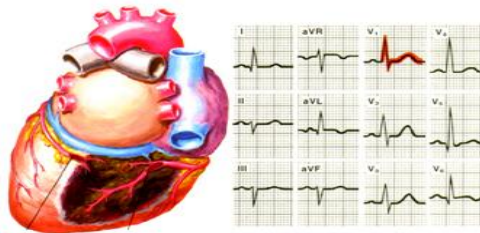
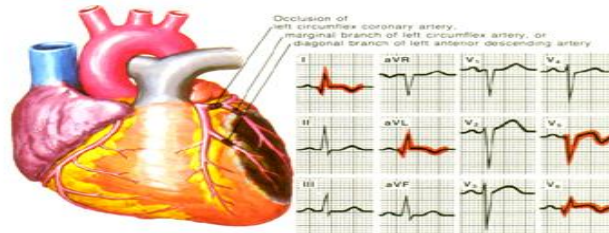
Assessment: the biblical character Nabal, who demonstrated features of hostile personality, became extremely angry because of the behavior of his wife Abigail. This anger led to a severe emotional upset that played a crucial role in the development of AMI. Since Nabal's past medical history was unremarkable, it is most likely that he was previously a healthy middle-aged man.

Myocardial Infarction

The clinical diagnosis of AMI relies on symptoms, electrocardiographic findings, and biochemical markers (troponin, serum creatine kinase, and creatine kinase-MB) (51-53). Acute ischemic syndromes are classified as unstable angina/non-ST-elevation MI (UA/NSTEMI) and acute ischemic syndromes with STEMI (51,52). The new diagnostic criteria and markers are leading to increased proportions of acute ischemic syndromes being recognized as acute MI (54). The diagnosis of acute MI relies on the combination of clinical and biochemical tools, each providing its own diagnostic contribution (51).

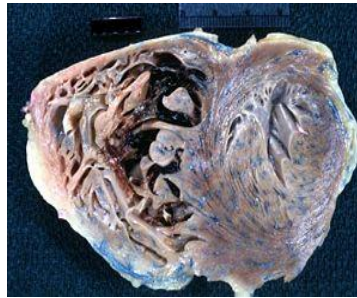


Acute myocardial infarction

Anterior MI**Posterior MI****Lateral MI**

The pathological hallmark of acute MI is a coagulative necrosis of the myocardium. In fatal MI, the pathological study must be performed at the appropriate technical and interpretative level to

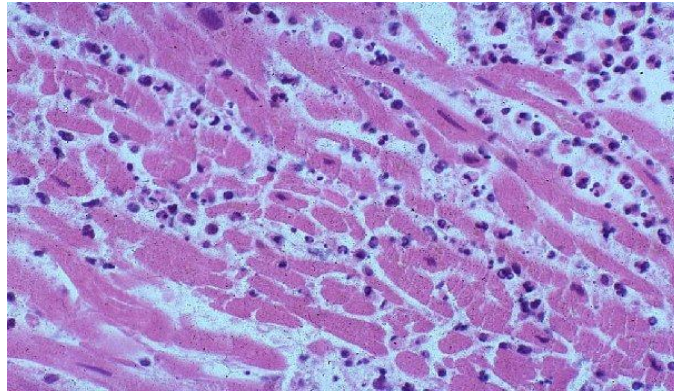
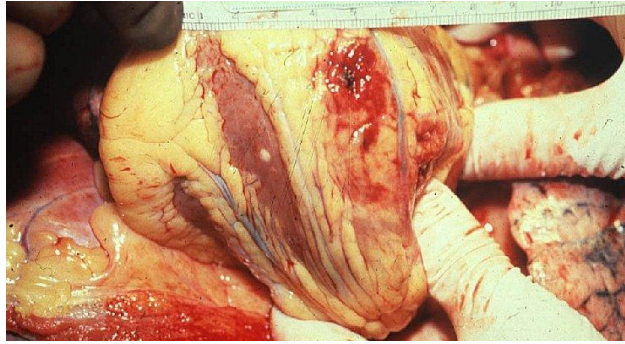
confirm, extend and improve information useful for the clinical understanding of the event (why one infarction is fatal while the other clinically similar MI is not) (55).



Myocardial Infarction

Most patients with acute MI who are admitted to coronary care units (CCUs) and coronary interventional labs shortly after the onset of the ischemia have a favorable prognosis (56). In the modern cardiology setting, fatal MIs are usually those occurring in patients out of the hospital, or in patients who came late to the CCU, or did not receive appropriate treatments, or died suddenly from life-threatening arrhythmias (57). With respect to transmural versus subendocardial MI, the identification of small intramural foci of coagulative necrosis, clinically recognized with the additional information derived from troponin measurements indicates the need for modified investigation protocols at autopsy with extensive search for microfoci of necrosis in multiple myocardial samples. These MIs are unlikely to be fatal unless the acute ischemia triggers life-threatening arrhythmias and, in any case, the corresponding clinical phenotype should be UA/NSTEMI (51).

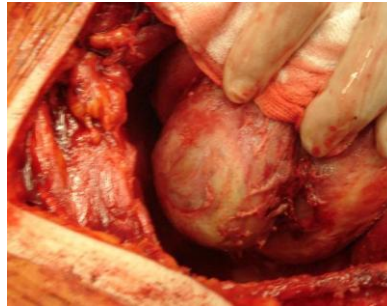
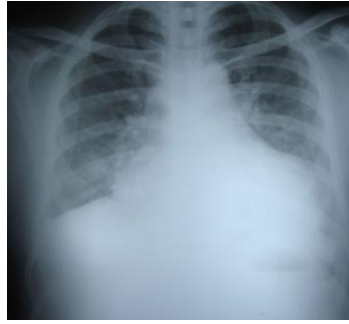
A culprit plaque with acute thrombosis is found at autopsy in more than 90% (51,52) of patients who have died from MI and were not treated with either thrombolysis or percutaneous transluminal coronary angioplasty (PTCA). The plaque substrate for thrombosis is a rupture in about 75% of the cases and erosion in a minority of cases (57), mostly in women and smokers (58). The typical culprit lesion is a large atherosclerotic plaque with cap ulceration and superimposed acute thrombosis. The acute thrombus is red, with a small platelet-rich small head, a fibrin- and red cell-rich body, and a red cell-rich tail (59).



Ruptured Myocardial Infarction

Complications of Acute Myocardial Infarction

Nabal died on about the 10th day after the onset of AMI, therefore it can be concluded that he died from some complication associated with AMI. What was the complication of MI that led to Nabal's death?



Left Ventricular Aneurysm

The major complications of AMI include: infarct extension, which is a progressive increase in myocardial necrosis within the infarct zone of the original AMI, occurring when an infarction extends and involves the adjacent myocardium or when subendocardial infarction becomes transmural (60), ventricular septum rupture (VSR) (61), LV free wall rupture (62,63), papillary muscle rupture (55), acute mitral regurgitation (MR) (64), cardiogenic shock (65), aneurysm in left anterior wall (66), pseudoaneurysm (67), CHF (68), pulmonary edema (69), dynamic LV obstruction (70), pericarditis (71), pericardial effusion (PE) (72), mural thrombosis (73,74), LV failure with cardiogenic shock (64), right ventricular (RV) failure, tamponade (75), hypertension, and tachyarrhythmias and bradyarrhythmias (76).

Was infarct extension and recurrent MI responsible for Nabal's death? Since infarct extension and recurrent MI are the long-term complications of AMI (77), it seems unlikely that extension or recurrent MI were responsible.

The incidence of VSR is 1-2% in patients with transmural infarctions, occurring mostly in the first week after infarction, and it remains an infrequent but devastating complication of AMI (60).

Cardiogenic shock before surgery, severe right heart failure, older age, female sex, anterior wall infarction, and no previous smoking predict death (32,78). In patients treated with fibrinolytic therapy, the highest risk is within the first 24 hours after MI. The rupture site can rapidly expand and cause sudden hemodynamic collapse, even in patients who appear to be clinically stable with normal LV function (78).

Was VSR responsible for Nabal's death? Nabal died on about the 10th day. Thus, although the possibility of VSR cannot be absolutely excluded it seems unlikely that VSR was responsible.

LV free wall rupture is an uncommon but catastrophic event following AMI, and is associated with high mortality (61,62). Cardiac rupture occurs in 1–6% of all patients admitted with STEMI. The frequency of cardiac rupture shows two peaks: one early within 24 hours, and one late from 3–5 days after STEMI (79,80). Cardiac rupture takes place after AMI in seven out of 15 (46.7%) consecutive patients within the first 24 hours (81), in 50% of patients within 5 days, and in 90% within 2 weeks (59). Risk factors for cardiac rupture include: first MI, anterior infarction, old age, female sex, hypertension during the acute phase of STEMI, lack of prior angina and MI, lack of collaterals, Q waves on the ECG, symptoms of pericarditis, MB creatine kinase >150 IU/l, intake of corticosteroids or non-steroidal anti-inflammatory drugs, and fibrinolytic therapy more than 14 hours after onset of symptoms. The most important determinants in preventing rupture are successful early reperfusion and the presence of collateral circulation (79,80).

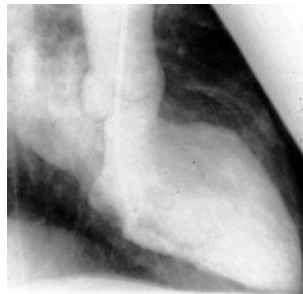
In Nabal's case, there is only one documented risk factor - the first MI. Since Nabal died on about the 10th day after AMI, this complication may have been responsible.

Papillary muscle rupture occurs in less than 1% of cases. The diagnosis is based on clinical and imaging findings (54). During the past 21 years, 13 patients came to autopsy (11 died in cardiogenic shock) with a papillary muscle rupture complicating AMI. Rupture occurred from 2 to 7 (mean, 4) days after the infarct, and survival after rupture was usually brief (median, 3 days). The infarct involved between 1% to 50% (mean, 19%) of patients, and in 10 it was less than 25% of the LV. In all instances, myocardium around the mitral annulus was not infarcted. Because papillary muscle rupture occurred mostly with first infarcts (eight), involved relatively small areas of necrosis, and spared the myocardium surrounding the annulus, early mitral valve replacement makes this cause of fatal AMI as one of the most treatable forms of cardiogenic shock (82).

A 32-year-old male was admitted with severe dyspnea. Blood examination indicated AMI. Echocardiogram revealed massive MR with prolapse of the anterior mitral leaflet due to rupture of the papillary muscle. Percutaneous coronary intervention for total occlusion in the right coronary artery was successfully performed, but progressive CHF continued to develop. Surgery for the papillary muscle rupture was performed on the 3rd day. Complete head rupture of the anterior papillary muscle was found and the mitral valve was replaced with a prosthetic valve (St. Jude Medical valve: #31). Pathological findings showed necrosis in the papillary muscle with inflammatory changes. The postoperative course was uneventful and the patient was discharged on the 43rd day after surgery (63).

Was a papillary muscle rupture responsible for Nabal's death? Nabal died on about the 10th day following AMI thus, this complication may have not been responsible for Nabal's death.

Acute MR, unless rapidly diagnosed and treated, is associated with high morbidity and mortality (83). Following AMI, MR of mild to moderate severity is found in 2-13%, while in 46% of patients, usually within the first two to seven days, and is responsible for 5% of deaths (84-86). After STEMI, MR may occur because of infarction of the papillary muscle, infarction involving the lateral wall, large infarction with LV dilation, and displacement/dysalignment of the papillary muscle. Severe MR with cardiogenic shock has a poor prognosis. In the SHOCK trial registry, approximately 10% of patients with shock presented with severe MR (overall hospital mortality 55%) (87). Cardiogenic shock developed on average after 12.8 hours in patients with MR (after AMI) (88). Post mortem MR was found in 7% of patients, with cardiogenic shock contributing to 5% of the mortality after AMI (89).



Mitral regurgitation

When severe MR is caused by infarction of the papillary muscle and wall, the area of infarction tends to be less extensive than in patients in whom the MR is caused by papillary displacement/dysalignment and severe LV dysfunction (54).

The presence of acute pulmonary edema or cardiogenic shock in posterior/posterolateral STEMI should point to the possibility of acute MR caused by papillary muscle rupture (64).

Was MR responsible for Nabal's death? Although MR as a cause of death cannot be absolutely excluded, it seems unlikely that it was responsible. Cardiogenic shock associated with MR can be ruled out since it usually develops within hours, but not after several days.

LV aneurysm usually occurs in the left anterior wall, in association with left anterior descending occlusion and a wide infarcted area. Patients with STEMI, treated with fibrinolytic therapy, and a patent infarct-related artery have a significantly reduced incidence of LV aneurysm compared with those who do not (7.2% vs. 18.8%) (65).

A study included 30 male patients with a mean age of 46 ± 7.6 years, with ST elevation of more than 1 mm on ECG, at least 15 days after the presumed onset of AMI. All patients were investigated by echocardiography and cardiac catheterization, and ST elevation was recorded. Aneurysm after STEMI in the ECG leads correlated with the site of necrosis. 2D echocardiography identified 21 cases (70%) of LV aneurysms and cardiac catheterization revealed 26 cases (86%); the electrical site of ST elevation correlated with the site of the aneurysm. This study indicated that the persistence ST elevation on the ECG more than 15 days after the onset of AMI has a good predictive value for the presence of LV aneurysm, which can be confirmed by cardiac catheterization (90).

Was LV aneurysm responsible for Nabal's death? Although this possibility cannot be excluded absolutely it seems that it was not responsible.

Ventricular pseudoaneurysm is a rare complication of AMI. It occurs as a consequence of rupture of the ventricular free wall and is contained by overlying, adherent pericardium, producing what has been termed a "false aneurysm or pseudoaneurysm" of the LV. The pathologic features depend on the interval of time elapsed from onset of MI and death and from the extent of hemorrhage between the pericardium and the myocardial wall. The myocardial wall shows interruption or fissuring. The myocardial changes include coagulative necrosis with or without reperfusion pattern, according to the administered treatments (66). Most pseudoaneurysms are formed

within seven days following AMI, only exceptionally developing later.

Was ventricular pseudoaneurysm responsible for Nabal's death? Since Nabal died on about the 10th day, it seems that this complication was not responsible.

Heart failure in patients with AMI is characterized either by diastolic dysfunction alone or by both systolic and diastolic dysfunction (91). LV diastolic dysfunction leads to pulmonary venous hypertension and pulmonary congestion, while systolic dysfunction is principally responsible for a depression of cardiac output and of the ejection fraction (92). More than 75% of patients with pulmonary congestion and AMI have an ejection fraction of less than 40% and more than a third have an ejection fraction of less than 30% (67). Pulmonary edema may occur as an acute event with the onset of STEMI or reinfarction, or as the culmination of slowly progressive CHF during the first days after MI, and is associated with a 20–40% 30-day mortality rate, even in the fibrinolytic era (68).

Was Nabal afflicted by CHF? Pulmonary edema? The biblical text describes: "*.. that his heart died within him..*". Although the likelihood of this complication is small, it is still possible that pulmonary edema led to his death.

Cardiogenic shock is the commonest cause of death in AMI. Although the syndrome of cardiogenic shock complicating AMI is common, the spectrum of underlying pathology is broad (92). The incidence of cardiogenic shock complicating AMI is 7.1% (93) with mortality rates between 50% to 80% (93-96). Patients with cardiogenic shock have a significantly greater risk of dying than those who do not develop cardiogenic shock (71.7% vs. 12%, $p < 0.001$) (93). Cardiogenic shock in patients with STEMI is commonly (75%) caused by extensive LV dysfunction (64). Other relevant causes include mechanical complications (acute severe MR, VSR, and subacute free-wall rupture with tamponade). Cardiogenic shock may be mimicked by aortic dissection and hemorrhagic shock (64, 97).

Most deaths occur within 48 hours from the onset of shock (95). Overall, shock develops on average 6.2 hours after AMI symptom onset (98), while late shock (≥ 24 hours) is associated with recurrent ischemia, Q waves in two or more leads and left anterior descendent culprit vessel (98).

A risk assessment prognostic algorithm of 30-day mortality included clinical and hemodynamic data prospectively collected among patients with cardiogenic shock in the 41,021-patient Global Utilization of Streptokinase and Tissue Plasminogen Activator for

Occluded Coronary Arteries (GUSTO-I) trial. Logistic regression modeling techniques were used to evaluate the relations between demographic, clinical, and hemodynamic characteristics and 30-day mortality rate for the entire shock population (n=2968) and for patients who underwent right-heart catheterization (n=995). The odds (95% CI) of dying were 1.49 times higher (1.27-1.74) for patients 10 years older and 1.70 times higher (1.19-2.44) for patients with prior infarction. Findings derived from physical examination, such as altered sensorium and cold, clammy skin, were important independent predictors of prognosis (odds of dying 1.68 times higher for each, 1.19-2.39 and 1.15-2.46). The odds of dying were also 2.25 times higher (1.61-3.15) in patients with oliguria. Mortality rate was lowest for cardiac output and pulmonary capillary wedge measurements of 5.1 L/min and 20 mm Hg, respectively, and increased with either higher or lower values. Patients with earlier onset cardiogenic shock had better outcomes than patients in whom shock developed later, although for the latter subgroup, prognosis was worse for patients who had heart failure (Killip class II to III) (99).

Between April 1993 and August 1997, 1,190 patients with cardiogenic shock were registered and 232 were randomized in the SHOCK Trial. Predominant LV failure (78.5%) was most common, with isolated RV failure in 2.8%, severe MR in 6.9%, VSR in 3.9% and tamponade in 1.4%. In-hospital Registry, mortality was 60%; VSR was associated with a significantly higher mortality (87.3%) than all other categories ($p<0.01$). The risk profile and mortality were lower for Registry patients who were managed with thrombolytic therapy and/or intra-aortic balloon counter-pulsation, coronary angiography, angioplasty and/or coronary artery bypass surgery. After adjusting for these differences, the extent to which survival was improved with early revascularization was similar to that observed in the randomized SHOCK Trial. In this prospective Registry, the etiology of cardiogenic shock was a mechanical complication in 12% of patients (89).

Was cardiogenic shock responsible for Nabal's death? Although the likelihood of this complication is small, it is still possible that a late onset shock led to his death.

Dynamic LV obstruction is an uncommon complication of AMI and is a mechanism that can lead to a severe heart failure, cardiogenic shock (101), a free wall rupture (102), and sudden death (100-102).

Was dynamic LV obstruction responsible for Nabal's death? Since dynamic LV obstruction occurs between 14 and 48 hours

following AMI (100), this complication can be eliminated as the cause of death in this ancient patient.

In AMI, mural thrombosis occurs in about 20% of patients (in 60% of patients with large anterior infarcts), while thromboembolism develops in 15-40% of patients (103). Mural thrombus is formed on the inflamed endocardium over the area of infarction, and its fragments can break off and embolise to various organs such as brain, spleen, kidney, gut, and lower limbs (77). The highest risk of systemic embolism is in the first 10 days, and persists for at least three months (103).

Before the widespread use of primary percutaneous coronary intervention (PCI) and glycoprotein IIb/IIIa inhibitors (GP IIb/IIIa) LV thrombus formation had been reported to complicate up to 20% of AMIs. Consecutive patients (n=92) with ST-elevation AMI treated with PCI and GP IIb/IIIa inhibitors underwent 2-D echocardiograms, with and without echo contrast agent, within 24–72 hours. Only 4/92 (4.3%) had an LV thrombus, representing a significantly lower incidence than that reported in the pre-PCI era. Use of contrast agents did not improve detection of LV thrombi in this study. The incidence of LV thrombus formation after acute MI, in the era of rapid reperfusion, was lower than previously reported (74).

LV mural thrombus was detected by echocardiography in 41 patients after AMI, who were followed for four years (104). Echocardiography revealed predominant mural type of thrombi (38 patients) and none showed mobility. Embolic events were observed in six patients and one patient died following embolic stroke (104).

Can mural thrombus be blamed for Nabal's death? In the absence of clinical signs of embolic complications such as stroke, limb ischemia, renal infarction or intestinal ischemia, this cause of sudden death seems unlikely.

Pericarditis occurs in transmural STEMI involving the full thickness of the myocardial wall to the epicardium. Patients with pericarditis have larger infarcts, lower ejection fraction and higher incidence of CHF (105). There are two forms of pericarditis associated with AMI (106). The first is post-infarction pericarditis, which develops usually within 12 hours to 10 days after a large transmural AMI (104). The second form is Dressler's syndrome, which occurs in 1% to 6% of patients, with peak onset at 7 to 11 weeks post-AMI at a range of 1 to 28 weeks (106). The Dressler syndrome (post-MI syndrome) has essentially disappeared in the reperfusion era (105).

The incidence of both early postinfarction pericarditis and Dressler's syndrome appears to be declining. Pericardial pain and pericardial friction rub define early postinfarction pericarditis and usually develop on day 2 or 3 after a transmural AMI. The clinical course is benign, and the prognosis of the patient is not altered by development of this complication. Asymptomatic pericardial effusions (PEs) do not require specific therapy nor do they absolutely contraindicate the use of anticoagulation as was previously thought. The post-myocardial infarction syndrome develops usually during the second or third week after acute MI, but it can develop as early as 24 hours and as late as several months after the MI. Whether this syndrome is the result of autosensitization to myocardial antigens released into the circulation during infarction remains uncertain. Alternative hypotheses for the causation of the syndrome include the release of blood in the pericardial space and that the syndrome represents a prolonged and exaggerated form of early postinfarction pericarditis. Clinically, post-myocardial infarction syndrome is manifested by fever, malaise, chest pain, and the presence of a pericardial and possibly pleuropericardial friction rub. PE is frequently large, and, rarely, cardiac tamponade may develop that requires pericardiocentesis (70).

PEs are found in 15% to 28% of patients presenting with the first AMI (71,107). PEs are associated with higher morbidity and mortality rates, in part because they occur more frequently with larger AMI that tends to be transmural. Patients with early small PE are more likely to develop later moderate to severe effusions. Two-thirds of patients with moderate to severe PE develop tamponade or free wall rupture (108).

The aim of this study was to assess the prevalence of PE in AMI and the different prognosis associated with distinct patterns of PE (anechoic/hypoechoic vs. hyperechoic effusion). Consecutive patients (n=585) admitted to the Coronary Care Unit for AMI were initially evaluated. Forty of them were excluded due to a technically poor acoustic window. The remaining 545 patients were studied by two-dimensional echocardiography at admission before discharge (after an average of 9 days in the Coronary Care Unit) and whenever there was an important change in the clinical status (chest pain, lipothymia or syncope, hemodynamic deterioration with systolic blood pressure <90 mmHg, or cardiac arrest). PE was found in 51 patients (9%). Three distinct textural patterns of PE were noted on the basis of the echogenic properties: 1) anechoic or hypoechoic PE was frequent (30 patients), mild or moderate and generally benign; 2) hyperechoic type

"A" effusion pattern was rare (2 patients) and was associated with fever, leukocytosis and pericardial rubs; 3) hyperechoic type "B" was frequent (19 patients), large and always associated with major complications (in all cases cardiac tamponade and/or death). PE was a common finding in serial echo evaluations of patients with AMI, especially when infarction was anterior, extensive and with Q wave. Echocardiographically detected PE showed different textural patterns; benign hypoanechoic effusion was more frequent, limited and benign than hyperechoic effusion, which was larger and was associated with adverse events (109).

Was Dressler syndrome responsible for Nabal's death? Nabal died on about the 10th day. Dressler syndrome occurs at 7 to 11 weeks, at a range of 1 to 28 weeks. Although this complication may have happened, a benign clinical course indicates that this syndrome was not responsible.



Dressler syndrome

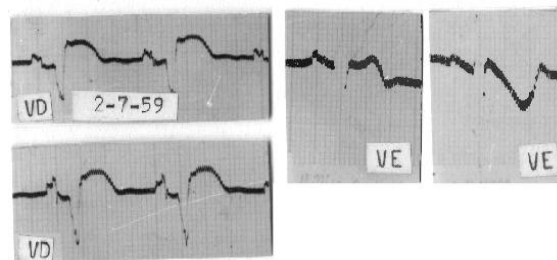
Was a post-infarction pericarditis responsible? This type of pericarditis develops usually within 12 hours to 10 days after a large transmural AMI. Since the clinical course of a post-infarction pericarditis is benign, it seems unlikely that this type of pericarditis afflicted Nabal.

Was Nabal afflicted by PE? PEs are associated with higher mortality rates, in part because they occur more frequently with larger AMI that tend to be transmural. Thus, it may be possible that Nabal suffered from this complication.

Right ventricular myocardial infarction (RVMI) is an independent risk factor for increased mortality even in these days of primary percutaneous coronary intervention (110). Isolated RVMI is a very rare complication of ischemic heart disease. Generally it accompanies infero-posterior or antero-septal MI cases. RVMI is a strong predictor of acute RV failure, bradyarrhythmia, ischemic and mechanical complications and is frequently complicated by cardiogenic shock which often leads to death. Acute RV ischemia and

RVMI are big problem particularly during and early post-operative cardiosurgery procedures. Atherosclerotic changes and heart blood flow disturbances predispose to RV ischemia or RVMI that occur more often in cardiosurgical patients, especially in early postoperative period. On the other hand, early intraoperative diagnosis and longer reperfusion period result in the correction of heart function and better prognosis (111).

RVMI as assessed by various diagnostic methods accompanies inferior-posterior wall MI in 30% to 50% of patients. Recognition of the syndrome of RVMI is important as it defines a significant clinical entity, which is associated with considerable immediate morbidity and mortality and has a well-delineated set of priorities for its management. Patients may clinically present with hypotension, elevated jugular venous pulse (JVP), and occasionally with shock, all in the presence of clear lung fields. The ST-segment elevation of ≥ 0.1 mV in the right precordial leads V4R is a readily available electrocardiographic sign used for diagnosis of RVMI. Other diagnostic approaches for assessing RVMI include echocardiography, radionuclide ventriculography, technetium pyrophosphate scanning, and hemodynamic measurements. The proper management of RVMI includes volume loading to maintain adequate RV preload, inotropic support, and maintenance of atrioventricular synchrony. Reperfusion therapy should be initiated at the earliest signs of RV dysfunction. Finally, complete recovery over a period of weeks to months is a rule in a majority of patients, suggesting that was RV "stunning" rather than irreversible necrosis (112).



Right ventricular myocardial infarction

RVMI complicates up to half of inferior LV infarctions. The term represents a spectrum of disease from asymptomatic, or mild RV dysfunction to cardiogenic shock, and it includes transient ischemic myocardial dysfunction as well as myocardial necrosis. RVMI is associated with considerable morbidity and mortality, and its presence

defines a high-risk subgroup of patients with inferior LV infarction. Diagnosis of this condition requires a high degree of suspicion based on clinical findings and the early recording of the electrocardiogram through right precordial leads, as well as elevated right-sided filling pressures out of proportion to left-sided filling pressures. The proper management of RVMI requires sustaining adequate RV preload with volume loading and maintenance of atrioventricular synchrony, reduction of RV afterload (particularly when LV dysfunction is present), and inotropic support of the RV. Early reperfusion with fibrinolytic therapy or direct angioplasty is also warranted. Survivors of RV infarction generally have a restoration of normal RV function with resolution of hemodynamic abnormalities (113).

Was RVI responsible for Nabal's death? We have insufficient data to answer this question.

A study determines the causes of cardiac tamponade, focusing especially on haemopericardium (HP), as a terminal mode of death, within a 430,000 rural English population. The hospital mortuary register and, all postmortem reports between 1995 and 2004, were interrogated for patients dying of cardiac tamponade or HP. The causes of cardiac tamponade/HP and selected morphological characteristics were then determined. Postmortems (n=14,368) were performed in this period: of these, 461 patients died of cardiac tamponade. Three cases were due to non-hemorrhagic PE. HP accounted for the remaining 458 cases of which, five were post-traumatic, 311 followed rupture of an AMI (RAMI), 138 developed intra-pericardial rupture of dissecting ascending aortic aneurysms (RD3A) and four were due to miscellaneous causes. HP was more commonly due to RAMI. Men tended to die from RAMI or RD3A earlier than women did. Two thirds of RAMI were associated with coronary artery thrombosis. Anterior free wall rupture was commonest in women, while posterior free wall rupture in men. The volume of intrapericardial blood in RAMI (mean = 440 ml) and RD3A (mean = 498 ml) varied between 150 and 1000 ml: intrapericardial blood volume was greater in men than in women dying from either RAMI or RD3A. At postmortem, cardiac tamponade was most often related to HP, attributable to either RAMI or intrapericardial RD3A. Post-traumatic and other causes of cardiac tamponade were infrequent (75).

Was cardiac tamponade due to rupture of AMI responsible for Nabal's death?

MI can occur without coronary arteriosclerosis, with normal coronary arteries. The numerous causes of such AMI include arteritis, trauma to coronary arteries, coronary mural thickening with metabolic

diseases, emboli to coronary arteries, congenital coronary artery anomalies, myocardial oxygen demand-supply disproportion, and luminal narrowing by other mechanisms such as spasm of coronary arteries (Prinzmetal's angina with normal coronary arteries) (114).

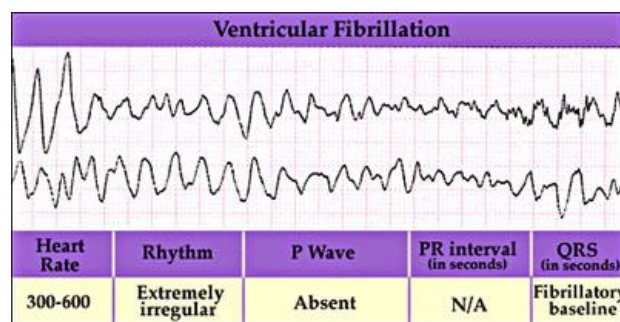
Nabal was a healthy middle-aged man. Therefore, among all the numerous causes listed above, it is most likely that in this case Prinzmetal's angina was responsible for the development of AMI.

Cardiac arrhythmias are common in patients with STEMI and occur most frequently early after the development of symptoms (115). Ventricular arrhythmias are one of the most frequent causes of death in non-hospitalized patients with acute MI. They are the most common form of sudden ischemic death (116).

The mechanisms for ventricular tachyarrhythmia include loss of transmembrane resting potential, re-entrant mechanisms due to dispersion of refractoriness in the border zones between infarcted and non-ischemic tissues, and the development of foci of enhanced automaticity. Reperfusion arrhythmias involve washout of toxic metabolites and various ions such as lactate and potassium (115).

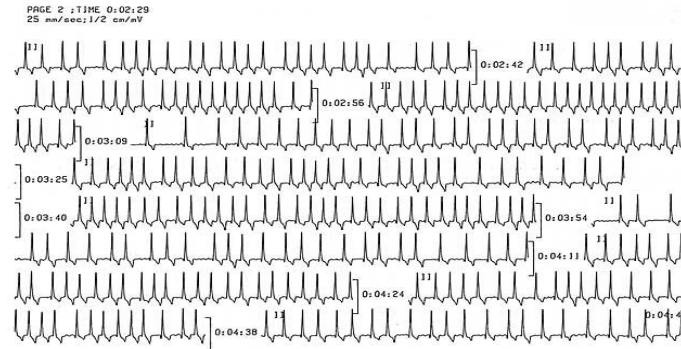
Dysrhythmia is related to the formation of re-entry circuits at the confluence of the necrotic and viable myocardium. PVCs occur in approximately 90% of patients. Non-sustained VT occurs in up to 49% of AMI patients while either sustained VT or ventricular fibrillation (VF) occur in about 3% to 5% (117).

The rate of VF is high in the first hour from the onset of pain and falls rapidly by six hours (118). VT and VF are major causes of morbidity and mortality after MI (119). Frequently, sudden cardiac death is the first manifestation of such malignant rhythms (120). Mortality in the immediate postinfarction period, as well as during the first year after AMI is most often due to sudden death from VF (121).



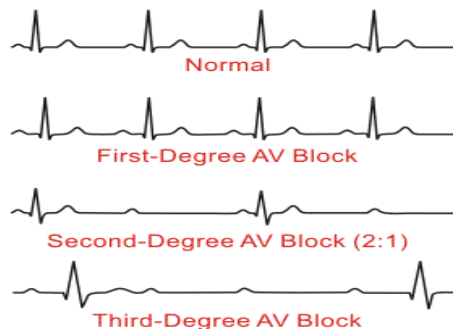
Atrial fibrillation (AF) is a common complication of AMI with a reported incidence as high as 21% (120) and is an independent

predictor of increased mortality following AMI (78,121-124). AF may exist before AMI or may develop following AMI due to ischemia of the sinus node or atrial myocardium, or sub-acutely following the development of CHF or pericarditis (123,125). Worsening health, hypotension, third degree atrial-ventricular (AV) heart block, VF, advanced age, a prior AMI, and coronary revascularization are independent predictors of new-onset AF (120,124).



Atrial fibrillation

Bradyarrhythmias, including AV block and sinus bradycardia, occur most frequently with inferior AMI. Complete AV block occurs in approximately 20% of patients with acute RVMI. Intranodal conduction disturbances with wide complex ventricular escape rhythms occur most frequently in large anterior AMI and have a very poor prognosis (59).



Were any of the types of arrhythmia, as mentioned above, responsible for Nabal's death? Acute emotional state may contribute of provoking VF and sudden death in susceptible patients even without AMI (126). The data on electrocardiographic evaluation are

unavailable in this case and any of the rhythm disturbances may have occurred. Nevertheless, it is more likely that the negative emotional stress that triggered AMI also provoked VF, which caused Nabal's death.

The foregoing description can improve the skills of contemporary physicians' in diagnosing and treating patients. We see that little has changed throughout the ages. Today as in ancient times, patients continue to be afflicted by AMI. The same clinical picture, the same complications, and the same causes of death. Both ancient and contemporary patients deserve adequate diagnostic evaluation and optimal treatment. In order to treat our patients properly, we need to study similar clinical events in our history.

In summary: biblical passages such as “...when the wine was gone out from Nabal, and his wife had told these things, that his heart died within him, and he became as a stone. And it came to pass about ten days after....he died” indicate that Nabal, a middle-aged man, was afflicted by AMI, with his death resulting from some complication of this event. Extreme anger superimposed on a very hostile personality led to severe mental stress that acted as a trigger in promoting AMI. The most likely mechanism of AMI in this case is Prinzmetal's angina, in which spasm of coronary arteries occurs. Among the numerous complications of AMI that may relate to Nabal's death are: infarct extension, VSR, LV wall rupture, papillary muscle rupture, MR, late cardiogenic shock, dynamic LV obstruction, aneurysm in left anterior wall, pseudoaneurism, CHF, pulmonary edema, pericarditis, PE, acute pulmonary hemorrhage, mural thrombosis, LV failure with cardiogenic shock, RVMI, RV failure, tamponade, and arrhythmias such as VT, VF, AF, complete AV block, and intranodular conduction disturbances with wide complex ventricular escape rhythms. Of these complications, the most likely are: LV free wall rupture, pulmonary edema, late cardiogenic shock, PE, which led to heart tamponade or rupture of cardiac muscle, or VF, provoked in Nabal's case by acute emotional stress.

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RESUSCITATION FROM ANCIENT TO MODERN TIMES

In medicine, resuscitation denotes therapeutic maneuvers intended to reverse acute, and life-threatening physiological abnormalities. The goal of resuscitation is to restore normal or near-normal cardiopulmonary function without deterioration of other organ systems (1). In popular culture, it connotes the restoration of life. This difference generates clinical and ethical dilemmas when the acronyms CPR (CardioPulmonary-Resuscitation) and DNR (Do-Not-Resuscitate) face each other over the deathbed (2).

CPR is an emergency first aid protocol for an unconscious patient in whom neither breathing nor pulse can be achieved (3). The medical term for heart stoppage in a patient is cardiac arrest, also referred to as cardiorespiratory arrest, in which case CPR is used. If the patient still has a pulse, but is not breathing, it is called respiratory arrest and rescue breathing is used (3).

Resuscitation medicine includes the pathophysiology and reversibility of acute terminal states and clinical death (4); its treatment methods, i.e., the 3 x 3 steps of the three phases of cardiopulmonary-cerebral resuscitation; its delivery systems, i.e., multidisciplinary critical care medicine (5-7) and community-wide emergency medical services (8), such as mobile and in-hospital intensive care units (9).

Resuscitation has developed over thousands of years, following a circuitous course of brilliant experimentation, serendipitous observation, slow adoption, and forgotten-then-rediscovered wisdom, which converged in the 1950s to bring about the era of modern CPR. The frontiers continue to be advanced in the quest to revive "hearts too good to die" (10). A rate of successful resuscitation remains disappointing, and aggressive dissemination of the current CPR guidelines should be encouraged because these evidence-based recommendations have been shaped by numerous observations and studies that share the common goal of improving resuscitation outcomes. Continued investigation is needed to assess the quality of standard CPR and to ensure adherence to performance guidelines as well as to explore ways to improve cardiac massage and preserve neurological function. A combination of novel techniques and broad implementation strategies will almost certainly yield further

improvement in efforts to resuscitate men and women who suffer from cardiopulmonary arrest.



Ancient Resuscitation

The correct techniques of resuscitation may restore individuals to life. Was resuscitation used in biblical times? If so, on whom? Who performed resuscitation? What were techniques used in resuscitation? Was resuscitation effective? Were resuscitation techniques different from those used today? In order to answer these questions (1) a quick historical look at resuscitation was taken, (2) all biblical texts were examined and cases of resuscitation were studied closely, and (3) contemporary techniques of resuscitation were described and compared with ancient accounts of resuscitation.

History of CardioPulmonary Resuscitation

Three and a half millennia (3.500 years) ago, in Egypt, the inversion method of resuscitation was described, involving hanging the patient by the feet, applying chest pressure to assist expiration, and releasing pressure to assist inspiration (1).

Already one millennium ago, it was recommended ventilating patients via bellows and a tube placed into the trachea (11).

During centuries, various other accounts of resuscitation included the use of flogging, bouncing on a trotting horse, rolling over a barrel, and sporadic descriptions of divine intervention (12).

Artificial respiration began in the 16th century with Vesalius's work on living animals introducing mechanical ventilation by bellows into the trachea (13).

1628. The possibility of resuscitating an arrested heart was described by William Harvey, who had observed restoration of the heartbeat in doves: "Trying an experiment on a dove, after the heart had quite left motion and that the ears [atrial appendages] had quite given over, I wetted my finger with spittle and being warmed kept it

awhile on the heart; the heart and its ears began to contract and did seem as if it were recalled back again from death" (14).

1732. Several cases of successful mouth-to-mouth resuscitation were the most famous of which was Tossach's report of the resuscitation of a clinically dead coal miner (15). He described the technique as "very simple and absolutely safe and can at least do no harm". The implications of this report were not much appreciated in the UK but in Holland, it attracted considerable attention, leading to the establishment of the first Humane Society for the resuscitation of the apparently drowned. Many such cases occurred in the Dutch canals, especially in icy winters. This new society advocated expired-air respiration, as did all the humane societies subsequently established all over Europe and on the east coast of the United States. The expiratory phase of respiration was encouraged by pressure on the chest and abdomen by the free hand (16,17).

1767. The Dutch Humane Society published guidelines for resuscitation of victims of drowning, stating: "keep the victim warm, give mouth-to-mouth ventilation, and perform insufflations of smoke of burning tobacco into the rectum" (1).

1775. The instructions provided by the English Humane Society in its first annual report included: "Artificial respiration [was] carried out in the following manner. The operator closed the patient's nostrils and applied his mouth to his and thus inflated the lungs and expanded the belly and the chest. By compressing these parts with his free hand the operator brought about an expiratory movement" (16,17).

1782. Ventilation by bellows was considered preferable to expired-air respiration and was adopted by all the humane societies. "Resuscitation sets" were placed at selected sites along the Thames for immediate use in cases of drowning. Inflation of the stomach was prevented by external compression of the esophagus in the neck, as had originally been advised by Hunter (and was later to become known as the Sellick maneuver for the prevention of esophageal reflux during induction of anesthesia) (18).

1788. Endotracheal intubation was introduced by Charles Kite, for which he was awarded a silver medal by the Royal Humane Society (18).

1792. External defibrillation had first been suggested by Curry, an advisor to the Royal Humane Society, following his experimental work on small animals he recorded two successful cases of human resuscitation by this means. He used two electrodes, one above the clavicle and the other over the lower left chest (19). Two years later a further case was reported in the Transactions of the Royal Humane

Society (20,21). A young child had fallen out of a window and was "to all appearances dead" when taken to hospital. Electric shocks from a friction type electrostatic electricity machine "transmitted through the thorax" successfully re-established the circulation and 'her health was restored'.

1800. Napoleon's personal physician Dominique Larrey made great strides in the care of traumatic injuries specifically to aid Napoleon's war efforts. He conceived the first "ambulance volantes", which made possible the rescue and resuscitation of wounded soldiers in battle, and led to the development of the modern ambulance service, as we know it today (22).

1868. John Hill of the Royal Free Hospital wrote the following description of sternal compression that it could equally be a present-day description of external cardiac massage. "The surgeon's left hand was placed firmly across the front of the chest, the fingers resting over the fifth, sixth and seventh costal cartilages on the right side, while the tip of the thumb laid on the second piece of the sternum and the muscular part of the hand on the corresponding cartilages on the left side. The right hand was now crossed over the left and forcible pressure made; the hands then being suddenly removed, the chest was allowed to expand by its own elasticity" (23).

1874. Moritz Schiff (1823–1896), professor of physiology in Florence, made fundamental observations on the cause and treatment of cardiac arrest while studying the cause of death under chloroform and ether anesthesia in animals. Noting that cardiac arrest preceded respiratory arrest he considered that treatment by chest compression, ventilation or the application of electricity was of no use. But "if one opens the thorax whilst air is slowly blown into the lungs and compresses the heart rhythmically with the hand to squeeze out the blood, compressing the abdominal aorta at the same time so that more of the artificial circulation is directed to the head, one can restore the heart as long as eleven and a half minutes after it has been arrested". He thus stressed the fundamental principle that the renewed activity of the heart was not merely the result of its mechanical irritation but also depended on an increased supply of oxygenated blood to the myocardium through the restored coronary circulation (24).

Following the work of Schiff, external cardiac compression was abandoned in the early twentieth century in favor of internal massage through an open chest and this continued to be used until the work of Kouwenhoven half a century later (25).

1878. The possibility that external compression of the heart without opening the chest would provide an adequate circulation was

first shown in cats by Boehm (Dorpat, Germany): "It can easily be seen that for a long time one could keep up a kind of emergency circulation; as soon as the compression is decreased the heart refills itself from the main veins, so that again and again fresh blood will be forced out of the heart" (26).

1899. Pré'vost and Batelli (Geneva) showed that electrically induced fibrillation could be reversed by a 240 volt alternating-current shock if applied to the heart within fifteen seconds. They also made the fundamental observation that, if a longer period elapsed, cardiac massage was necessary to provide myocardial oxygenation (27). However, successful internal and external defibrillation was not achieved until half a century later.

1901. Attempts to restore cardiac activity in man by internal massage were made in the late nineteenth century and success was first obtained by Igelsrud (Tromsø, Norway) (reported by Keen). After a cardiac arrest towards the end of an abdominal operation, the heart was exposed by resection of parts of the fourth and fifth ribs, the pericardium was opened and the heart was massaged "between the thumb on one side and the fore and middle fingers on the other". Pulsation returned after one minute and the patient made a complete recovery (28).

1902. A second case of cardiac activity restoration was reported. During an appendectomy by Arbuthnot Lane (London) under ether anaesthesia, cardiac and respiratory arrest had occurred simultaneously. Artificial respiration was performed and the heart started again "after a squeeze or two" from below the diaphragm (29).

1903. External cardiac compression was confirmed in dogs by George Crile in Cleveland (30).

1904. External cardiac compression was used in seven human cases in USA, "one successfully and two partially so" (30).

1909. Massage had been performed through the open chest or the abdomen, either through an incision in the diaphragm in the midline to open the pericardium or through the intact diaphragm from below. Cases (n=48) had been reported and success had been achieved in ten (31).

1933. William Kouwenhoven (1886–1975) rediscovered external cardiac compression by accident during his research on internal and external defibrillation, and so became the founder of modern CPR. By 1933, his team had developed satisfactory systems for both internal and external defibrillation (32). The greatest advance in the treatment of cardiac arrest since the time of Schiff was made when he demonstrated on a dog's fibrillating heart that an adequate

circulation could be maintained for up to thirty minutes by external compression of the chest wall.

1947. Internal defibrillation was used successfully by Beck (Cleveland) during an operation for correction of a funnel chest deformity. In this dramatic case, cardiac massage had been performed for thirty-five minutes, during ten of which the heart had been in ventricular fibrillation (VF) (33).

1953. The first successful internal defibrillation was performed on a patient with acute coronary occlusion who had developed VF. The chest was opened without sterile precautions and defibrillation was performed after a preliminary period of cardiac massage (34).

1954. Brock (London) reported a series of thirty patients in whom VF had developed during the course of a cardiac operation. Defibrillation was performed in twenty and nine of these recovered. The importance of preliminary cardiac massage was again emphasized (as recommended by Schiff 80 years earlier) (35).

1956. A second successful defibrillation was reported by Beck on a physician who collapsed while unwisely leaving hospital after electrocardiography had shown an early posterolateral infarct. VF was confirmed when the chest was opened in the nearby emergency operating room and the "heart squeezed intermittently against the sternum". There was no response to two shocks so the pericardium was opened and defibrillation was obtained after three further shocks (36).

1956. Zoll published his accounts of defibrillation (37).

1957. Safer Peter, "Father of CPR", documented the ABC (Airway, Breathing, and Circulation) of cardiopulmonary resuscitation and later coined the expression "cardiopulmonary cerebral resuscitation" with the acronym CPR (38).

1958. Safer and Elam formally presented mouth-to-mouth ventilation (39).

1958. The American Red Cross adopted mouth-to-mouth breathing as the preferred method of artificial respiration (40).

1960. Mobile intensive care unit ambulances, staffed by physicians, were created. Dr. Pantridge demonstrated that these mobile intensive care units could be used successfully by medical personnel to resuscitate outpatients with sudden cardiac death after acute myocardial infarction (22,41).

1960. Kouwenhoven, Jude, and Knickerbocker rediscovered external chest compression while testing defibrillation on an animal model of VF. The paper, describing this discovery (of chest

compressions used to resuscitate), was widely attributed as being the usher of the modern era of CPR (32).

1962. The American Heart Association proposed the term cardiopulmonary resuscitation (CPR) (42).

1973. The American Red Cross and the American Heart Association began a campaign to teach the American population CPR (43).

1991. The "chain of survival" concept, promoting bystander CPR, as well as early call to the ambulance service, early defibrillation, and early advanced care were promulgated (44).

This is a quick look at resuscitation techniques. Various techniques have been used throughout human history from rectal fumigation to the modern CPR and automated machines. We see that methods of resuscitation have changed constantly through the ages.

The Standard Treatment

CPR is now the standard treatment for someone who has a cardiac arrest. However, a procedure has emerged only relatively recently. For a number of scientific and religious reasons, it was long considered impossible, even blasphemous, to attempt to reverse "death". Because of these factors, the area of resuscitation failed to progress until the Enlightenment in the 18th Century. The main elements of resuscitation were then developed over the next 200 years, and eventually brought together to create CPR in the early 1960s. The increased demands, in which morality was placed on the medical profession to combat sudden cardiac death subsequent to this development, have been an important factor in why CPR has come to be used in the widespread manner it currently is (45).



The development of modern CPR has given every person the ability to challenge death anywhere. Despite sparks of knowledge and occasional applications of possibly effective lifesaving efforts since

antiquity, the possibility to reverse acute terminal states or clinical death by modern, physiologically sound, and effective measures did not come about until around 1900 inside hospitals, and around 1960 outside hospitals. Additional potentially effective cerebral resuscitation research since around 1970 was taken to clinical trials before the year 2000. The history of resuscitation medicine around 1900, when many opportunities to assemble existing bits of knowledge into an effective system were missed, should be a warning for those individuals who will lead CPR beyond the year 2000. History has shown the need for continuing communication and collaboration among investigators of different countries, and between laboratory researchers, clinicians of various disciplines, and prehospital rescuers. The lessons learned from history, for research challenges in the near future, include: a) the development of ultra-advanced life support to be initiated outside the hospital, to bridge CPR-resistant cases to definitive cardiac procedures in the hospital; and b) cerebral resuscitation to complete recovery after 10 to 15 min of normothermic cardiac arrest without blood flow. Both challenges above will require research projects at multiple levels - from the molecular and cellular levels, to the use of small and large animal models (with "organs and organisms" process and outcome evaluations), to studies of patients and communities. Beyond the year 2000, resuscitation research might become more challenging and cost-effective in the area of multiple trauma, which concerns the young and fit (46).

CPR is one of the most effective methods in rescuing those in critical situations. In recent decades, electric cardiac defibrillation has made the biggest advance in the field of CPR. The rate of successful resuscitation with electric defibrillation is much higher than by bare-handed or drug resuscitations, by this method more lives have been saved, and it has become the most essential and most important means of first aid. The history of the development of electric defibrillation is a successful modality of multidisciplinary cooperation of physicians, biologists, physiologists, and engineers. Although "early defibrillation" has been recognized as an idea of standard therapy and a basic measure of life support by international organizations as American Heart Association, it is far from being perfect and has a long way to go. A review of the history may help to bring the technique of electric defibrillation into perfection, and to save more lives in the future (47).

History grows, and while we look back at how we got to this stage in cardiac arrest resuscitation, new things come upon us. Two of

the more significant defibrillation developments in recent years have been the development of readily available automated external defibrillators (AEDs) and biphasic defibrillators. AEDs deliver defibrillating capacity without the need to interpret the cardiac rhythm. They are compact, easy to use, accurate, and are getting cheaper to buy. The distribution of this capacity to a variety of health professionals, and to lay resuscitators in some contexts (for example, at major sporting events, airports, etc - i.e., so called "public access defibrillation"), is an issue we will entertain over the next years (11).

Biphasic defibrillators, compared to traditional monophasic defibrillators, defibrillate at lower energy levels. However, not all biphasic defibrillators are the same, and the evidence base for the energy sequences to run through during resuscitation is not robust. Furthermore, the difference in machines and the uncertainty about energy doses has the capacity to confuse and distract operators during resuscitation (11).

After failure of initial external defibrillation, restoration of spontaneous circulation is largely contingent on rapid and effective reversal of myocardial ischemia by both mechanical and pharmacologic means. Despite the introduction of modern CPR more than 35 years ago, its universal acceptance, and its wide implementation, no improvements in outcome excepting early defibrillation have been documented over these many years. The science of CPR therefore is still in its infancy. It was incorrectly assumed that all that needs to be known is known and that the need for scientific research was therefore not apparent. Accordingly, serious resuscitation research was neither encouraged nor equitably supported. The ABCs of CPR currently provide for the establishment of a patent airway (A) and intermittent positive pressure ventilation, preferably with oxygen-enriched air; (B). These are to be immediately followed with precordial compression; (C). This ordering of priorities, however, is based on consensus rather than objective outcome measurements. The ABCs recently have been seriously challenged based on results of both experimental and clinical studies. Conventional external precordial compression restores systemic blood flow. Both professional and nonprofessional CPR providers, especially bystanders, because of its apparent simplicity and noninvasiveness may use it. However, manual or mechanical external precordial compression typically generates cardiac outputs that represent less than 30% of normal values. Coronary blood flow, which is critical for restoration of spontaneous circulation, is correspondingly reduced. Accordingly, several alternatives to conventional precordial

compression have been proposed with the intent of increasing cardiac output and both coronary and cerebral blood flows. Among the large number of pharmaceutical agents initially recommended for cardiac resuscitation, only agents that produce peripheral vasoconstriction are of proved benefit. Epinephrine has been the preferred vasopressor agent for the management of cardiac arrest for more than 35 years because of its alpha-adrenergic effects. However, the potentially adverse effects of epinephrine are related to its beta-adrenergic inotropic actions. The beta-adrenergic actions account for disproportionate increases in myocardial oxygen consumption with increased severity of myocardial ischemic injury and provocation of ectopic ventricular tachycardia (VT) and VF. Nevertheless, epinephrine remains the drug of choice, although adrenergic drugs with selective alpha-adrenergic actions or nonadrenergic vasoconstrictor drugs are likely to emerge as useful alternatives. Experimental and clinical observations have led to identification of continuous monitoring of both end-tidal carbon dioxide and VF waveforms as practical noninvasive guides because they are highly correlated with both cardiac output and coronary blood flow. Both end-tidal carbon dioxide and VF waveforms serve as predictors of the likelihood of successful resuscitation. These two measurements are used to guide interventions and especially to assure optimal precordial compression. Sudden death among adults is predominantly due to malignant ventricular arrhythmias and VF. Early defibrillation serves as an unequivocally effective immediate intervention. Minimally trained first responders and members of the public are being enfranchised to use automated external defibrillators for very early defibrillation. Use of these devices by bystanders is the most promising new intervention since CPR was first proposed in the early 1960s. Postresuscitation ventricular dysrhythmias and heart failure are now called postresuscitation myocardial dysfunction. This complication has been recognized as a leading cause of the high postresuscitation mortality (48).



Biblical Accounts of Resuscitation

The first case relates to a Shumanite's son who was revived by the prophet Elisha 896 BC: "*And he went up, and lay upon the child, and put his mouth upon his mouth, and his eyes, and put his hands upon his hands; and he stretched himself upon the child; and the flesh of the child waxed warm*" (II Kings 4:34). The second child concerns a son of the widow Zarephath who was revived by the prophet Elija 868 BC: "...*and he laid him (the son) upon his bed*" (I Kings 17:19). "*And he stretched himself upon the child three times...*", "...*and the soul of the child came into him again, and he revived*" (17:21,22). These accounts suggest resuscitation techniques performed on two children.

There is another hint of resuscitation in the Bible. When "...*the king of Egypt spake to the Hebrew midwives, the name of the one was Shiphrath, and the name of the other Puah. ...If it be a son, then you shall kill him: but if it be a daughter, then she shall live. But the midwivessaved (tehaen) the men children alive*" (Exodus 1:15-17). In The Holy Scriptures, English version, the Hebrew word *tehaen* is translated into English as saved (49). According to the Complete Hebrew English Dictionary (50), the Hebrew word *tehiah* is revival, rebirth, renewal or resurrection. It follows that midwives saved male-newborns (in defiance of Pharaoh's cruel decree) and carried out resuscitation on those newborn males who required this procedure. So long ago, before the prophets Elisha and Elija resuscitated children, it seems that resuscitation was carried out on newborn males and similarly on newborn females, if required. Since there is no description of how these newborn males were revived, or resuscitated, this research deals only with two children - the Shumanite's son and the son of the widow Zarephath.

According to an arbitrary definition, an "infant" refers to a child in the first year of life (51), while the "child" is aged from 1 year to about 8 years (52). The biblical text states clearly that both prophets, Elisha and Elija, revived children. Thus, according to the modern definition, these children were between 1 to 8 years. It follows therefore that biblical accounts of resuscitation are dealing with pediatric patients.

What were the causes of cardiorespiratory arrest that led to the resuscitation of the two biblical children?

Pediatric Deaths

Cardiopulmonary Arrest

The major causes of death in infants and children are injuries, sudden infant death syndrome (SIDS), sepsis, neurological diseases, and respiratory failure (53).

Injuries

Injuries are the leading cause of death in children and young adults, causing more childhood deaths than all other causes combined (53). Many injuries are preventable. The most common fatal childhood injuries amenable to prevention are motor vehicle passenger injuries, pedestrian injuries, bicycle injuries, drowning, burns, and firearm injuries (54).

Motor Vehicle Injuries

Motor vehicle related injuries account for nearly half of all pediatric deaths in the United States (53). Contributing factors include failure to use passenger restraints, inexperienced adolescent drivers, alcohol, not wearing a seat belt, speeding, and aggressive driving (55).

Bicycle Injuries

Bicycle crashes are responsible for about 20,000 injuries and nearly 150 deaths per year in children and adolescents (56). Head injuries are a major cause of bicycle-related morbidity and mortality. It is estimated that bicycle helmets can reduce the severity of head injuries by >90% (57).

Burns

Approximately 80% of fire-related and burn-related deaths result from house fires and smoke inhalation (58,59). Smoke detectors are the most effective way to prevent deaths and injuries; 70% of deaths occur in homes without functioning smoke alarms (60).

Firearm Injuries

The highest number of firearm deaths occurs in adolescents and young adults, with firearm injuries as more likely to be fatal in young children (61). The presence of a gun in the home is associated with an

increased likelihood of adolescent and adult suicides or homicides (62-64).

Sudden Infant Death Syndrome

SIDS is "the sudden death of an infant under 1 year of age, which remains unexplained after a thorough case investigation, including performance of a complete autopsy, examination of the death scene, and review of the clinical history" (65). The etiology of SIDS remains unknown, but risk factors include prone sleeping position, sleeping on a soft surface (66-68), and second hand smoking (69,70).

Drowning

Drowning is the second major cause of death from unintentional injury in children <5 years of age and the third major cause of death in adolescents. Most young children drown after falling into swimming pools while unsupervised; adolescents more commonly drown in lakes and rivers while swimming or boating (71).

As we see, there are numerous causes of cardio-respiratory arrests in children. Although they cannot be excluded absolutely, insufficient information makes the presence of any of these causes unlikely in the two cases studied in this research.

Cardiorespiratory Arrest

In general, prehospital primary cardiac arrest is a less common etiology in children and young adults than in older adults (72-74), with primary respiratory arrest being a more common etiology than primary cardiac arrest in children (75-79). In a 15-year retrospective study of prehospital cardiac arrest in the United States, 7% of 10,992 victims of prehospital cardiac arrest were younger than 30 years, and 3.7% were younger than 8 years (72), while only 2% of victims of in-hospital CPR in Great Britain were between 0 to 14 years of age (80).

Cardiac arrest in children is rarely sudden; it is typically the end result of deterioration in respiratory function or shock, and the terminal rhythm is typically bradycardia with progression to pulseless electrical activity or asystole (81,82). VT and VF have been reported in 15% or less of a subset of pediatric and adolescent victims of prehospital cardiac arrests (72,73), even when rhythm is assessed by the first responders (83,84). Survival following prehospital cardiopulmonary arrest averages approximately 3% to 17% in most

studies, and survivors are often neurologically devastated (73,75,77,81,83-89).

The two biblical children were not hospitalized, and resuscitation took place outside hospital. Was cardiac arrest responsible for the condition requiring resuscitation? Respiratory? Cardiopulmonary? It seems that in the case of Shumanite's son some cardiorespiratory cause was responsible, while in the son of Zarephath cardiac arrest occurred.



Modern Resuscitation

Evaluation of Biblical Resuscitation

Historical reports of CPR began with the biblical accounts of resuscitation by the prophet Elijah (90) and the prophet Elisha (91-93). In the case of the Shumanite's son the words "...put his mouth upon his mouth," indicates that the child had stopped breathing, while "... he stretched himself upon the child;" shows chest compression following cardiac arrest. In the case of the son of Zarephath, the words "And he stretched himself upon the child.." indicate chest compression, demonstrating that the child was breathing but his heart had stopped. According to Parascos (94) the terms used in the description are *crouching over* (another term used in translation), *mouth-to-mouth*, and *stretching himself* upon. Variant readings of these accounts include "breathing into" and "pressing upon" the afflicted youths. In these two accounts, there are strong suggestions of "artificial resuscitation" (94).

Step one of Elijah's method was that with his own body weight, he crushed the child's chest and abdomen and squeezed "out the bad air" (95). Step 2 was that by then rising, he allowed air to enter, to the extent that the rib cage recoiled. Alternatively, massive compression of the child's chest may have activated the excito-inspiratory Hering-Breuer reflex (95).

Elijah's feat is rarely mentioned in modern articles on mouth-to-mouth breathing, but it should be because Elijah's earlier case as

reported was really the control for his later “mouth-to-mouth” resuscitation. The reports of resuscitation by Elijah and Elisha are almost identical, as though a modern statistician had designed a perfect experiment: in each case, there was a widow and her dead son, in each the prophet placed the dead boy in his bed in the loft, in each the prophet stretched himself on the boy and the boy recovered. As in any perfectly controlled experiment, there was only one variable; in the control case, Elijah’s mouth made no contact with the boy’s, but in the experimental study, Elijah’s mouth did. The experiment failed because both patients recovered (95).

The Dynamics of Cardiopulmonary Resuscitation

During CPR, the dynamics of chest compression play a major role in determining outcome (96,97). Heart action and respiratory effort are involved in transporting oxygen to the tissues. The main organ to suffer from oxygen starvation is the brain, which may sustain damage after four minutes and irreversible damage after about seven minutes (3). The heart rapidly loses the ability to maintain a normal rhythm. Following cardiac arrest, effective CPR enables enough oxygen to reach the brain to delay brain death, and allows the heart to respond to defibrillation attempts (3). The best way to preserve neurological function is through early and effective restoration of cardiopulmonary function (28).

Rapid and effective CPR is associated with successful return of spontaneous and neurologically intact survival in children (75,97). The greatest impact occurs in respiratory arrest (98), in which neurologically intact survival rates of >70% are possible (77,86,99), while in VF survival rates of 30% have been documented (73). Only 2% to 10% of all children who develop out-of-hospital cardiac arrest survive, and most are neurologically devastated (73,83,87,88,100-103). Part of the disparity is due because bystander CPR is available for less than half of the victims of out-of-hospital arrest (99,100,102). Some studies show that survival and neurological outcome can be improved with prompt CPR (98,99,104,105).

If CPR is begun within several minutes of cardiac arrest, and defibrillation arrives shortly thereafter, a patient’s chances of recovery rise to approximately 80%. CPR is almost never effective if started more than 15 minutes after collapse because permanent brain damage has probably already occurred. An exception is cardiac arrest occurring with exposure to very cold temperatures, because hypothermia seems to protect the victim by slowing down metabolic

and physiological processes, greatly decreasing the oxygen demand by tissues (3).

In light of what the foregoing, we see that the two biblical children received prompt and effective resuscitation, leading to their survival, without leaving any neurological damage.

Steps in Resuscitation

CPCR for basic life support consists of steps A (airway support) (106), B (breathing support) (106,107), and C (circulation support) (108,109). Advanced life support, for restoration of spontaneous circulation, consists of steps D (drugs and fluids) (110-112), E (electrocardiography) (113), F (fibrillation treatment, electric defibrillation) (114-117), G for gauged (titrated) (46), H for humanized (brain-oriented), and I for intensive care (46).

What steps of resuscitation were carried out on the biblical children? Were all the above steps performed? We learn that in one case resuscitation was carried out using the following steps: B - Breathing and C - Circulation, while on the other only C - Circulation. These resuscitations indicate that even in ancient times basic knowledge of the principles of resuscitation in children existed. The results were rewarding: although only two steps were performed on the first child and one step on the second child, both children were resuscitated successfully and returned to life.

New Guidelines of CRP

Rescue Breathing and CPR

New guidelines (52) of the American Heart Association (November 2005) for CRP and Emergency Cardiovascular Care of Pediatric and Neonatal Patients: Pediatric Basic Life Support include:

Check for Response

Tap or gently shake the person or shout "Are you okay?" Call the child's name if you know it. If the child is responsive, he or she will answer or move. Quickly check to see if the child has any injuries or needs medical assistance. If necessary, leave the child to phone emergency medical services, but return quickly and recheck the child's condition frequently.

If the child is unresponsive and is not moving, shout for help and start CPR. If you are alone, continue CRP for 5 cycles (about 2 minutes). One cycle of CPR for the lone rescuer is 30 compressions

and 2 breaths. If a second rescuer is present, that rescuer should immediately activate the emergency medical services system. If you suspect trauma, the second rescuer may assist by stabilizing the child's spine. If the child must be moved for safety reasons, support the head and body to minimize turning, bending, or twisting of the head and neck.



Open the Airway

In an unresponsive infant or child, the tongue may obstruct the airway, so the rescuer should open the airway (118-121). If you are a lay rescuer, open the airway using a head tilt-chin lift maneuver for both injured and noninjured victims. The jaw thrust is no longer recommended for lay rescuers because it is difficult to learn and perform, is often an ineffective way to open the airway, and may cause spinal movement.

Check Breathing

Look for rhythmic chest and abdominal movement, listen for exhaled breath sounds at the nose and mouth, and feel for exhaled air on your cheek. Periodic gasping, also called agonal gasps, is not breath (122,123). If the child is breathing and there is no evidence of trauma, turn the child onto the side. This helps maintain a patent airway and decreases the risk of aspiration.

Look, listen, and feel for normal breathing for 5 to 10 seconds. If there is no breathing, go to the next step.

Rescue Breathing

If the child is not breathing or has only occasional gasps for the lay rescuer: maintain an open airway and give two breaths. For the healthcare provider: maintain an open airway and give two breaths.

Make sure that the breaths are effective (i.e., the chest rises). If the chest does not rise, reposition the head, make a better seal, and try again. In an infant, use mouth-to-mouth-and-nose technique; in a child, use a mouth-to-mouth technique (124).

Check Pulse

Try to palpate a pulse (brachial in infant and carotid or femoral in a child). Take no more than 10 seconds. If you do not definitely feel a pulse within 10 seconds, proceed with chest compressions.

Chest Compressions

Compress the lower half of the sternum but do not compress over the xiphoid. After each compression allow the chest to recoil fully because complete reexpansion improves blood flow into the brain (125). Characteristics of good compression are: "push hard": push with sufficient force to depress the chest approximately one third to one half of the anterior-posterior diameter of the chest; "push fast": push at a rate of approximately 100 compressions per minute; release completely to allow the chest to fully recoil; minimize interruptions in chest compressions.

In an infant victim, lay rescuers and lone rescuers should compress the sternum with two fingers placed just below the intermammary line. The two thumb-encircling hands technique is recommended for healthcare providers when two rescuers are present. Encircle the infant's chest with both hands; spread your fingers around the thorax, and place your thumbs together over the lower half of the sternum (126-130). Forcefully compress the sternum with your thumbs as you squeeze the thorax with your fingers for counter pressure (131-132).

In a child, lay rescuers and health care providers should compress the lower half of the sternum with the heel of one hand or two hands (as used for adult victims) but should not press on the xiphoid or the ribs. The chest should be compressed about one third to one-half the anterior-posterior depth of the chest.

For lay rescuers, a single compression-ventilation ratio (30:2) for all age groups may increase the number of bystanders who perform CPR because it is easier to remember.

If you are the only rescuer, perform cycles of 30 chest compressions followed by 2 effective ventilations with as short a pause in chest compressions as possible. Make sure to open the airway before giving ventilations. For 2-rescuer CPR, one provider should perform chest compressions while the other maintains the airway and

performs ventilations at a ratio of 15:2 with as short a pause in compressions as possible. The 15:2 ratios for two rescuers are applicable in children up to the start of puberty. The compressing rescuer should deliver 100 compressions per minute continuously without pauses for ventilation. The rescuer delivering the ventilations should give 8 to 10 breaths per minute and should be careful to avoid delivering an excessive number of ventilations.



Several Changes

The 2005 CPR and Emergency Cardiovascular Care (ECC) guidelines include several key changes from the previous version published in 2000. The new guidelines place an increased emphasis on chest compressions and recommend a compression: ventilation (C:V) ratio of 30:2. Current knowledge on defibrillation has also been incorporated by recommending that Emergency Medical Service (EMS) rescuers give two minutes of CPR before defibrillation when the response interval is greater than four to five minutes and EMS responders did not witness the arrest. Another major change is the recommendation for a single shock to be administered followed immediately by CPR with no check of the cardiac rhythm until two minutes of CPR has been performed postdefibrillation. An automated external defibrillator should be implemented in public locations where there is a relatively high likelihood of witnessed cardiac arrest. In addition, the most recent guidelines highlight the shift from primary-rhythm-based therapies and resuscitation to a focus on neurologic outcomes. Several evidence-based changes were included in the 2005 CPR and ECC guidelines, including a C:V ratio of 30:2 and mitigation of hands-off time, early defibrillation, administration of a single shock versus a three-shock sequence, use of public-access defibrillators, and a shift from primary-rhythm-based therapies to a focus on neurologic outcomes (133).

Basic life support should be started as early as possible. Lay rescuers should not check for a pulse, they should call for help and start chest compressions immediately. Compression depth should be

4-5 cm at a rate of 100 compressions per minute, with chest compressions and ventilation in a ratio of 30:2, as stated previously. Lay rescuers should continue until professional help arrives. Lay rescuers may use the same procedure for children as recommended for adults. Professionals should, however, initiate CPR in children with 5 ventilations followed by a compression-ventilation ratio of 15:2. Automatic External Defibrillation should be used as early as possible (134).

During CPR with an advanced airway in place, rescuers will no longer perform "cycles" of CPR. Instead, the rescuer performing chest compressions will perform them continuously at a rate of 100/minute without pauses for ventilation. The rescuer providing ventilation will deliver 8 to 10 breaths per minute (1 breath approximately every 6-8 seconds). Timing of 1 shock, CPR, and drug administration during pulseless arrest has changed and now is identical to that for advanced cardiac life support. Routine use of high-dose epinephrine is not recommended. Lidocaine is de-emphasized, but it can be used for treatment of VF or pulseless VT if amiodarone is not available. Induced hypothermia (32-34 degrees C for 12-24 hours) may be considered if the child remains comatose after resuscitation. Intact survival has been reported following prolonged resuscitation and absence of spontaneous circulation despite 2 doses of epinephrine. The following are the major neonatal resuscitation changes in the 2005 guidelines: supplementary oxygen is recommended whenever positive-pressure ventilation is indicated for resuscitation; free-flow oxygen should be administered to infants who are breathing but have central cyanosis. Although the standard approach to resuscitation is to use 100% oxygen, it is reasonable to begin resuscitation with an oxygen concentration of less than 100% or to start with no supplementary oxygen (i.e., start with room air). If the clinician begins resuscitation with room air, it is recommended that supplementary oxygen be available to use if there is no appreciable improvement within 90 seconds after birth. In situations where supplementary oxygen is not readily available, positive-pressure ventilation should be administered with room air. Current recommendations no longer advise routine intrapartum oropharyngeal and nasopharyngeal suctioning for infants born to mothers with meconium staining of amniotic fluid. Endotracheal suctioning for infants who are not vigorous should be performed immediately after birth. A self-inflating bag, a flow-inflating bag, or a T-piece (a valved mechanical device designed to regulate pressure and limit flow) can be used to ventilate a newborn. An increase in heart rate is the primary sign of improved

ventilation during resuscitation. Exhaled CO₂ detection is the recommended primary technique to confirm correct endotracheal tube placement when a prompt increase in heart rate does not occur after intubation. The recommended intravenous (IV) epinephrine dose is 0.01 to 0.03 mg/kg per dose. Higher IV doses are not recommended, and IV administration is the preferred route. Although access is being obtained, administration of a higher dose (up to 0.1 mg/kg) through the endotracheal tube may be considered. It is possible to identify conditions associated with high mortality and poor outcome in which withholding resuscitative efforts may be considered reasonable, particularly when there has been the opportunity for parental agreement. The following guidelines must be interpreted according to current regional outcomes: When gestation, birth weight, or congenital anomalies are associated with almost certain early death and when unacceptably high morbidity is likely among the rare survivors, resuscitation is not indicated. In conditions associated with a high rate of survival and acceptable morbidity, resuscitation is nearly always indicated. In conditions associated with uncertain prognosis in which survival is borderline, the morbidity rate is relatively high, and the anticipated burden to the child is high, parental desires concerning initiation of resuscitation should be supported. Infants without signs of life (no heartbeat and no respiratory effort) after 10 minutes of resuscitation show either a high mortality rate or severe neurodevelopmental disability. After 10 minutes of continuous and adequate resuscitative efforts, discontinuation of resuscitation may be justified if there are no signs of life (135,136).

Last guidelines include significant modification in the management of cardiac arrest patient. Recognition of cardiac arrest by lay rescuers is performed on the absence of vital sign (no reactivity, no breathing) and it is now only recommended for healthcare providers to check the pulse. A short period of CPR before attempting defibrillation may be considered in adults with out-of-hospital VF or pulseless VT and a delay to EMS response. Defibrillation is provided with biphasic waveform at 150-200 J and is immediately followed by a 2 min period of CPR. Adrenaline remains the drug of choice in cardiac arrest whatever the first rhythm because no benefit has been demonstrated with vasopressin in term of survival. Amiodarone is the first line antiarrhythmic, which improving short-term survival, is recommended after the second shock for resistant VF. Postresuscitation treatment is receiving greater emphasis in emergency cardiovascular care, but there is little evidence to support specific therapies with the exception of hypothermia (12-24 hours at

32-34 degrees C) that is currently recommended if patient remains unconscious after VF. Revascularization should also be discussed if cardiac arrest is presumed to be from ischemic origin (136).

In a nationwide, prospective, population-based, observational study, 5170 children aged 17 years and younger who had an out-of-hospital cardiac arrest, from Jan 1 2005 to Dec 31 2007, were enrolled. Data collected included age, cause, and presence and type of CPR by bystander. The primary endpoint was favorable neurological outcome 1 month after an out-of-hospital cardiac arrest, defined as Glasgow-Pittsburgh cerebral performance category 1 or 2. Children (n=3675, 71%) had arrests of non-cardiac causes and 1495 (29%) cardiac causes. 1551 (30%) received conventional CPR and 888 (17%) compression-only CPR. Data for type of CPR by bystander were unavailable for 12 children. Children who were given CPR by a bystander had a significantly higher rate of favorable neurological outcome than did those not given CPR (4.5% [110/2439] vs. 1.9% [53/2719]; adjusted odds ratio [OR] 2.59, 95% CI 1.81-3.71). In children aged 1-17 years who had arrests of non-cardiac causes, favorable neurological outcome was more common after bystander CPR than no CPR (5.1% [51/1004] vs. 1.5% [20/1293]; OR 4.17, 2.37-7.32). However, conventional CPR produced more favorable neurological outcome than did compression-only CPR (7.2% [45/624] vs. 1.6% [six of 380]; OR 5.54, 2.52-16.99). In children aged 1-17 years who had arrests of cardiac causes, favorable neurological outcome was more common after bystander CPR than no CPR (9.5% [42/440] vs. 4.1% [14/339]; OR 2.21, 1.08-4.54), and did not differ between conventional and compression-only CPR (9.9% [28/282] vs. 8.9% [14/158]; OR 1.20, 0.55-2.66). In infants (aged <1 year), outcomes were uniformly poor (1.7% [36/2082] with favorable neurological outcome). For children who have out-of-hospital cardiac arrests from non-cardiac causes, conventional CPR (with rescue breathing) by bystander is the preferable approach to resuscitation. For arrests of cardiac causes, either conventional or compression-only, CPR is similarly effective (137).

We see that techniques of resuscitation have changed little throughout the ages. The biblical accounts referred to here are most likely the first description of resuscitation carried out on children.

Summary

Various methods of resuscitation have been used through the ages as therapeutic procedures to treat acute and life threatening situations. This research examines whether there are biblical descriptions of resuscitation. All biblical texts were examined and two accounts of children who were revived were studied closely. The first was a Shumanite's son who was revived by the prophet Elisha: *"And he went up, and lay upon the child, and put his mouth upon his mouth, and his eyes, and put his hands upon his hands; and he stretched himself upon the child; and the flesh of the child waxed warm"*. The second was a son of the widow of Zarephath who was revived by the prophet Elija: *"...and he laid him (the son) upon his bed, And he stretched himself upon the child three times,...and the soul of the child came into him again, and he revived"*. The techniques of revival indicate resuscitation, which includes in the case of the first child B - Breathing and C - Chest Compression, while in the second child: C - Chest Compression. We see that during long human history the basic principles of CPR are described in the Bible, although new more efficient methods were introduced and ineffective procedures were dropped. These biblical accounts of resuscitation are most probably the first descriptions resuscitation in human history. It can be concluded that the roots of modern CPR go back to biblical times.

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CONCLUDING REMARKS

Cardiovascular diseases have afflicted humans since the dawn of history. Even in contemporary days, many diseases have not disappeared from the scene. Biblical text were examined and the verses relating to the four cardiovascular diseases were studied closely.

This research analyzes the disease that afflicted the Biblical King Asa (the third of the King of the house of Judah, reigned between 867-906 BCE). The elderly have suffered from pain in their legs, which may be associated with various diseases, for thousands of years. The sentence *"Nevertheless in the time of his old age he was diseased in his legs"* indicates that King Asa suffered from disease in his legs. Among numerous diseases, peripheral artery disease, gout, and degenerative osteoarthritis were most likely to affect the King's legs. Among these diseases, the diagnosis of peripheral artery disease is the most acceptable.

The second part deals with Jacob, the son of Isaac and Rebecca and the third of the Patriarchs of the Jewish people. After Jacob's sons left Egypt, they returned to their father, Jacob, and told him that their brother: *"Joseph is still alive, and he is governor over the lands of Egypt. And Jacob's heart fainted, for he believed them not"* (Genesis 45:26). We see that Jacob fainted, and a transient syncope was responsible for this situation. Among various causes of syncope that could have affected Jacob, the most likely are vasovagal syncope or some cardiac arrhythmia.

Subsequently, the research examines from a contemporary perspective the biblical description of a patient who died from some type of a heart disease. Passages such as *"...when the wine was gone out from Nabal, and his wife (Abigail) had told these things, that his heart died within him, and he became as a stone. And it came to pass about ten days after....he died"* demonstrate an acute heart disease, a condition that indicates acute myocardial infarction (AMI), with death resulting from some complication of this acute event. Nabal's hostile personality in addition to extreme anger and mental stress, associated with his wife's deceitful behavior, triggered the development of the infarction.

The most likely mechanism of AMI in this case is Prinzmetal's angina, in which spasm of coronary arteries occurs. Among the numerous complications of AMI that may relate to Nabal's death are: infarct extension, ventricular septal rupture, left ventricular (LV) wall rupture, papillary muscle rupture, late cardiogenic shock, dynamic LV

obstruction, aneurysm in left anterior wall, pseudoaneurism, congestive heart failure, pulmonary edema, pericarditis, pericardial effusion, mural thrombosis, LV failure with cardiogenic shock, right ventricular failure, tamponade, and arrhythmias such as ventricular tachycardia, ventricular fibrillation (VF), atrial fibrillation, complete A-V block, and intranodular conduction disturbances with wide complex ventricular escape rhythms. Of these complications, the most likely are: LV free wall rupture, papillary muscle rupture, pulmonary edema, late cardiogenic shock, pericardial effusion, which led to heart tamponade or rupture of cardiac muscle, or VF, provoked in Nabal's case by acute emotional stress. This is probably the first description of a middle-aged man who died from AMI, triggered by various psychosocial factors.

Various methods of resuscitation have been used through the ages as therapeutic procedures to treat acute and life threatening situations. These include: the inversion method - hanging the patient by the feet, applying chest pressure to assist expiration and releasing pressure to assist inspiration, flogging, bouncing on a trotting horse, rolling over a barrel, a divine intervention, mouth-to-mouth resuscitation, external chest compression, and ABC (Airway, Breathing, and Circulation) technique, as well as the usage of semi-automatic defibrillator, and automatic cardiopulmonary resuscitation machine. This research studies ancient descriptions of resuscitation, looks at its history and presents a modern approach towards its basic steps. All relevant biblical texts were examined and the cases of two children who were revived by the prophets were studied. The first was a Shumanite's son who was revived by the prophet Elisha: *"And he went up, and lay upon the child, and put his mouth upon his mouth, and his eyes, and put his hands upon his hands; and he stretched himself upon the child; and the flesh of the child waxed warm"*. The second was a son of the widow of Zarephath who was revived by the prophet Elija: *"...and he laid him (the child) upon his bed, And he stretched himself upon the child three times,...and the soul of the child came into him again, and he revived"*. From the point of view of contemporary medicine, the process of revival indicates resuscitation in the case of the first child B - Breathing and C- Chest Compression, while in the second child: C – Chest Compression. These two biblical accounts are most probably the first descriptions of resuscitation in human history. It can be therefore concluded that the roots of resuscitation can be traced back to biblical times.